

AN EXPLICIT FINITELY PRESENTED NON-MF GROUP

ANONYMOUS

ABSTRACT. A countable discrete group is MF in the Carrión–Dadlart–Eckhardt–Shulman sense if it embeds into the unitary group of a norm matrix corona $\prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$. As of August 2026, whether every group is MF remained open. The precise group notion dates to 2012–2013 and has an operator-algebraic lineage going back to the Blackadar–Kirchberg MF program of 1997. It is one of a family of metric approximation problems whose normalized-Hamming (sofic) case was solved recently by OpenAI and subsequently generalized by Kun and Thom, and whose unnormalized Frobenius case was solved by De Chiffre, Glebsky, Lubotzky and Thom. We construct an explicit finitely presented group E , together with a designated central involution $w \in E$, such that $w \neq 1$ but *every* homomorphism from E to the unitary group of every norm matrix corona of the displayed form kills w . Consequently E is not MF, and neither its full nor its reduced group C^* -algebra is an MF algebra. The group is built from $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, one HNN letter implementing the proper endomorphism $(v, A) \mapsto (2v, A)$, and one additional involution; the proof combines the Kazhdan projection of Γ with stable finiteness of matrix coronas. The finite-dimensional case of the kill theorem is elementary and uses neither property (T) nor unitarity; we also show, via quasidiagonality of amenable groups, that the same marked presentation with a cyclic base can have a corona representation in which the marked involution survives. The kill theorem is proved as a general finite-normal criterion: if a countable group contains the image of a property (T) group, compressed by a letter t and centralized by a letter c , then every finite normal subgroup contained in the compression-defect subgroup — the normal closure of the commutators of tct^{-1} with that image — is killed by every homomorphism into the unitary group of every C^* -subalgebra of a norm matrix corona. Among the consequences: $C_{\mathrm{red}}^*(E)$ is a unital separable C^* -algebra with a faithful tracial state — hence stably finite — that is not an MF algebra, giving an explicit reduced-group- C^* -algebra realization of the already known failure of stable finiteness to imply the MF property; MF groups are closed under neither quotients nor split extensions; every countable group has a largest MF quotient, by its MF radical, and the radical element w is portable: any group receiving E with w surviving is non-MF; for every $m \geq 2$ a scaled variant E_m ($E = E_2$) is a finitely presented non-MF group, and there exist $2^{\aleph_0}$ pairwise non-isomorphic finitely generated non-MF groups; and the kill theorem holds in a uniform quantitative form over all matrix sizes at once.

1. INTRODUCTION

Let $(d_n)_{n \in \mathbb{N}}$ be a sequence of positive integers and let

$$(1) \quad \mathcal{Q} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$$

be the associated *norm matrix corona*: bounded sequences of matrices modulo sequences converging to 0 in operator norm. Following Blackadar–Kirchberg [BK], a separable C^* -algebra is *MF* if it embeds into some algebra of the form (1). We use *MF group* exclusively in the sense introduced by Carrión–Dadarlat–Eckhardt and used by Shulman [CDE, Sh]: a countable discrete group E is MF if there is an *injective* group homomorphism

$$E \longrightarrow \mathcal{U}(\mathcal{Q})$$

for some choice of (d_n) . Equivalently, E admits finite-dimensional approximate unitary representations that are approximately injective, with all defects measured in operator norm. This qualification matters: Shulman notes that a strictly stronger group property has also been called MF [Sh, Definition 3 and footnote 1]: *purely* MF requires the approximating maps to be genuine finite-dimensional representations. The two properties must not be conflated when comparing with the literature: Magee–de la Salle proved that $\mathrm{SL}_4(\mathbb{Z})$ is not purely MF [MdIS], while $\mathrm{SL}_4(\mathbb{Z})$ is residually finite and hence MF in the sense used here. Residually finite groups are MF, amenable groups are MF as a consequence of the Tikuisis–White–Winter theorem [TWW], free products of amenable groups are MF, and a substantial permanence theory exists [Sh].

The age of the problem has two layers. Blackadar and Kirchberg introduced MF C^* -algebras in 1997 [BK]; the broader stably-finite-to-MF embedding question attributed to Kirchberg implies a positive answer to the corresponding operator-norm approximation problem for groups [DGLT, Introduction]. The precise group property used here was introduced as Definition 2.7 by Carrión–Dadarlat–Eckhardt in a preprint first posted on October 15, 2012 and published in 2013 [CDE]. Their paper introduces the definition but does not formulate the universal group question in those words, so we distinguish the 1997 problem lineage from the 2012–2013 definition. At the date of the present result, these were respectively about twenty-nine years and nearly fourteen years old. As late as May 2026, Bachner–Dogon–Lubotzky described whether every group is MF as a longstanding open problem [BDL, pp. 235–238]. It is the operator-norm member of the family of approximation problems surveyed in [DGLT, BDL], whose unnormalized-Frobenius case was solved negatively in [DGLT]. Building on rigidity theorems for sofic approximations of Kazhdan groups due to Kun [Ku16] and Kun–Thom [KT19], OpenAI constructed the first nonsofic group [OAI]; Kun and Thom then extracted and generalized the mechanism, obtaining nonsofic generalized wreath products [KT], and Fournier–Facio exhibited finitely presented torsion-free examples [FF]. In the sofic problem the metric is normalized Hamming

distance on permutation groups, not rank distance; the three metric conventions are stated together in [Sh, Introduction]. Neither result supplies an operator-norm conclusion: [DGLT] treats the unnormalized Frobenius norm, while permutation/Hamming-metric arguments do not directly control operator-norm defects.

Against this August 2026 literature record, the theorem below answers the question negatively with an explicit finitely presented example.

Theorem A (non-MF group). *Let $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, let $\alpha: \Gamma \rightarrow \Gamma$ be the injective endomorphism $\alpha(v, A) = (2v, A)$, and let $a = (e_1, 1) \in \Gamma$, which does not lie in $\alpha(\Gamma)$. Let E be the finitely presented group obtained from Γ by adjoining a stable letter t with $t\gamma t^{-1} = \alpha(\gamma)$ for $\gamma \in \Gamma$, an involution c commuting with Γ , and by imposing that*

$$w := [tct^{-1}, a(tct^{-1})a^{-1}]$$

is central and satisfies $w^2 = 1$ (Definition 3.2). Then:

- (1) *$w \neq 1$ in E , so w is a central involution;*
- (2) *for every sequence (d_n) and every group homomorphism $\Theta: E \rightarrow \mathcal{U}(\mathcal{Q})$ into the corresponding norm matrix corona (1), one has $\Theta(w) = 1$.*

In particular E is not MF, and neither $C_{\max}^(E)$ nor $C_{\mathrm{red}}^*(E)$ is an MF C^* -algebra.*

A completely explicit presentation of E — eight generators, with every relation displayed — appears as (5) in Definition 3.2.

The proof turns a C^* -algebraic finiteness argument into a group-level kernel theorem. For orientation, it can be organized into the following five steps. The individual ingredients and the surrounding compression/lamp architecture have substantial antecedents. What is new here is the specific stable-finite Kazhdan pinning argument in a norm matrix corona and its conversion into a universal marked kernel.

- (1) *Central corner.* If $\Theta(w) \neq 1$, then $e = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection commuting with the image of Θ ; cutting by a lift of e replaces $\mathcal{Q}$ by another norm matrix corona on which $\Theta(w) = -1$ (Lemma 6.4). This cut eliminates small-relative-rank phenomena: on the corner, the deviation of w from 1 has full normalized size. This is a norm-corona adaptation of the negative-eigenspace compression for a central involution in the proof of [SV, Proposition 2.7]; the distinction is that no positive relative-rank estimate is needed here.
- (2) *Conjugation representation.* Unitary lifts $U_{g,n}$ of the corner representation conjugate the corner coordinates $M_{r_n}(\mathbb{C})$, and exact multiplicativity of Θ in the quotient makes arbitrary coordinate lifts asymptotically multiplicative in operator norm. Thus $g \mapsto (\mathrm{Ad} U_{g,n})_n$ defines an *exact* homomorphism β into the norm corona of the algebras $B(L^2(M_{r_n}))$ (Lemma 7.4).
- (3) *Kazhdan projection.* Γ has property (T), so its maximal group C^* -algebra contains the Kazhdan projection p_Γ , whose image under every

representation is the projection onto the invariant vectors. The HNN relation $t\Gamma t^{-1} = \alpha(\Gamma) \subseteq \Gamma$ forces, in *every* C^* -representation, the order relation $P \leq \beta(t)P\beta(t)^*$ between the images of p_Γ and its t -conjugate (Lemma 7.5).

- (4) *Stable finiteness.* A norm matrix corona is stably finite: isometries in every matrix amplification are unitaries (Lemma 6.2). Hence the two unitarily equivalent projections in step (3) must be equal: $\beta(t)$ preserves the Γ -fixed space.
- (5) *The marked word.* The involution c commutes with Γ , so the matrix vector representing $\Theta(c)$ is Γ -fixed; by step (4) so is the vector representing $\Theta(tct^{-1})$; hence $a \in \Gamma$ asymptotically fixes it, and the commutator word w is forced to 1 in normalized Hilbert–Schmidt norm along the chosen ultrafilter — while the corner normalization of step (1) says it equals -1 in operator norm. The two conclusions contradict each other (Section 8).

Nontriviality of w is witnessed by a completely explicit representation of E into a semidirect product of a Clifford group by a linear group over $\mathbb{Z}[1/2]$ (Section 4); no combinatorial theory of HNN extensions is needed.

Two companion results clarify the mechanism. First, in finite dimensions the kill theorem is elementary and needs neither property (T), nor unitarity, nor centrality of w :

Theorem B (finite-dimensional obstruction). *Let k be a field, V a finite-dimensional k -vector space, H a group, $\Gamma \leq H$ a subgroup, and $t, c \in H$, $a \in \Gamma$ elements with $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi: H \rightarrow \text{GL}(V)$ satisfies*

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = 1.$$

Applied to $H = E$, Theorem B shows that *every* finite-dimensional representation of E over *every* field kills w : finite-dimensional linear representations do not separate the points of E , no finite quotient can witness $w \neq 1$, and the infinite Clifford witness of Section 4 cannot be replaced by a finite-dimensional one. Property (T) enters the corona argument for one reason: it makes “the projection onto the Γ -fixed space” an element of the corona, uniformly across all coordinates at once.

Second, the fixed-space rigidity in that argument cannot simply be omitted:

Theorem C (cyclic-base comparison). *Let $G_0 = \text{BS}(1, 2) = \langle \gamma_0, t \mid t\gamma_0 t^{-1} = \gamma_0^2 \rangle$, let $\Gamma_0 = \langle \gamma_0 \rangle \cong \mathbb{Z}$, let $X_0 = G_0/\Gamma_0$, and let $E_0 = \text{CLamp}(X_0) \rtimes G_0$ be the corresponding Clifford extension (Construction 4.1), with central sign z_0 . Also define the literal marked-presentation analogue*

$$E_{\text{BS}} = \left\langle \gamma_0, t, c \left| \begin{array}{l} t\gamma_0 t^{-1} = \gamma_0^2, \quad c^2 = 1, \quad [c, \gamma_0] = 1, \\ w_{\text{BS}}^2 = 1, \quad [w_{\text{BS}}, \gamma_0] = [w_{\text{BS}}, t] = [w_{\text{BS}}, c] = 1 \end{array} \right. \right\rangle,$$

where $w_{\text{BS}} = [tct^{-1}, \gamma_0(tct^{-1})\gamma_0^{-1}]$. Then E_0 is an amenable MF group and $z_0 \neq 1$. There is a quotient homomorphism $E_{\text{BS}} \twoheadrightarrow E_0$ sending w_{BS} to z_0 ;

consequently the literal analogous presentation has a corona representation in which its marked involution survives. Nevertheless, every finite-dimensional representation of either group kills the corresponding marked element.

The quotient statement makes the comparison at the level actually used by the proof: the marked-word relations and witness mechanism survive when the Kazhdan base is replaced by $\mathbb{Z}$, but the universal corona-kill conclusion does not. Thus property (T), or a substitute furnishing a uniform fixed-space projection, is essential to this argument; this is not a claim that every possible proof of the concrete Theorem A must invoke property (T).

Generalizations and consequences. The corona half of Theorem A is proved once, in the following generality (Definition 7.1). A *marked Kazhdan pattern* in a countable group H is a tuple (Λ, ι, t, c) consisting of a countable group Λ with property (T), a homomorphism $\iota: \Lambda \rightarrow H$ — injectivity is *not* assumed — and elements $t, c \in H$ with $t\iota(\Lambda)t^{-1} \subseteq \iota(\Lambda)$ and $[c, \iota(\Lambda)] = 1$. Writing $d = tct^{-1}$, the *compression-defect subgroup* of the pattern is the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H.$$

The *finite-normal obstruction criterion* (Theorem 7.2) states: every finite normal subgroup of H contained in N_{comp} is killed by every corona representation of H . Theorem A(2) is the instance $H = E$ with $F = \{1, w\}$: since $c^2 = 1$ forces $d^2 = 1$, the marked word satisfies the identity $w = [d, a]^2 \in N_{\text{comp}}$ (Lemma 8.1). The corner cut that starts the proof is the finite-group averaging (Reynolds) projection $|F|^{-1} \sum_{f \in F} \Theta(f)$, whose complement for $F = \{1, w\}$ is exactly the central corner of step (1) above. Corollary 9.4 upgrades the target from coronas to arbitrary C^* -subalgebras of coronas, so that, for instance, every homomorphism from H into the unitary group of a unital separable quasidiagonal C^* -algebra, or of $C_{\text{red}}^*(F_k)$ for a free group F_k [HT], kills such an F . Theorem C calibrates the hypotheses: for the same pattern over the base $\mathbb{Z}$, the conclusion fails.

Section 9 records consequences in four directions.

(i) *C^* -algebras.* Blackadar and Kirchberg proved that MF algebras are stably finite and asked about the converse: whether every stably finite separable C^* -algebra is MF [BK], a question sometimes described — in Kirchberg’s stronger formulation, a conjecture — as a C^* -analogue of the Connes embedding problem (see, e.g., the introduction of [Ra]). The converse was already known to fail after the negative solution of the Connes embedding problem from $\text{MIP}^* = \text{RE}$ [JNVWY]: Goldbring gives the direct deduction by passing from a non- $\mathcal{R}^\omega$ -embeddable II_1 factor to a separable stably finite non-MF C^* -algebra [Go23, Corollary 5.2]. The new point here is that the counterexample is the reduced group C^* -algebra of the explicit finitely presented group constructed above. Indeed, as recently as March 2026 no full or reduced group C^* -algebra was known not to be MF [Sh, Introduction]:

Theorem D (a stably finite non-MF reduced group algebra). $C_{\text{red}}^*(E)$ is a unital separable C^* -algebra which carries a faithful tracial state — and is therefore stably finite — but is not an MF algebra.

Here $C_{\text{red}}^*(E)$ is not nuclear (E contains an isomorphic copy of Γ by Corollary 4.5, and Γ is not amenable), so the quasidiagonality problems for nuclear stably finite algebras around [BK, TWW] are untouched (Remark 9.7). In the opposite direction, for the ambient HNN extension G of (4) the algebra $C_{\text{max}}^*(G)$ contains an explicit proper isometry — so it is not finite, in particular not MF — although the group G itself is MF, being residually finite by the Borisov–Sapir mapping-torus theorem [BoS] (Proposition 9.9).

(ii) *Permanence.* MF groups are closed under neither quotients nor split extensions: E is a quotient of the MF group F_8 , and the witness group $W = \text{CILamp}(\bar{X}) \rtimes \bar{G}$ of Section 4 is a finitely generated non-MF group that is a split extension of the residually finite linear group $\bar{G}$ by the locally finite group $\text{CILamp}(\bar{X})$ (Corollaries 9.11 and 9.13); both statements contrast with the positive permanence theory of [Sh].

(iii) *Structure.* The MF radical of a countable group — the intersection of the kernels of its corona representations, cf. the finite and finite-approximate quotients of [Slo17] — is functorial under arbitrary homomorphisms, and the quotient by it is the largest MF quotient, realized by a single corona representation (Lemma 9.2 and Proposition 9.3; the structural statement is presumably folklore, and we claim no novelty for it). Portability of the radical element w has a striking consequence: embedding E into a 2-generator simple group S , which is possible by Goryushkin’s theorem [Go], gives a 2-generator simple group all of whose homomorphisms into MF groups are trivial (Corollary 9.14).

(iv) *Multiplicity and uniformity.* For every $m \geq 2$ the construction with the endomorphism $(v, A) \mapsto (mv, A)$ yields a finitely presented non-MF group E_m with $E = E_2$ (Theorem 9.15); the groups $E \times \mathbb{Z}^k$, $k \in \mathbb{N}$, are pairwise non-isomorphic finitely presented non-MF groups, and there exist $2^{\aleph_0}$ pairwise non-isomorphic finitely generated non-MF groups (Corollary 9.18). Finally, the kill theorem is uniform over all matrix sizes at once (Theorem 9.19): for every $\varepsilon > 0$ there are $\delta > 0$ and a finite $F_0 \subseteq E$ such that every map $\varphi: E \rightarrow \mathcal{U}(M_r(\mathbb{C}))$, for any r , with $\|\varphi(g)\varphi(h) - \varphi(gh)\| \leq \delta$ for all $g, h \in F_0$, satisfies $\|\varphi(w) - 1\| < \varepsilon$; the proof is a compactness reduction to Theorem 7.2 and yields no explicit pair (δ, F_0) .

What this paper does not prove. The proof of Theorem A does *not* show that E is nonhyperlinear; equivalently, it does not show that the canonical group trace, or the associated group von Neumann algebra, fails the relevant Connes-embedding property. It also does not touch the question whether hyperlinear groups are sofic. The finiteness step (4) is special to operator-norm coronas: in a tracial (Hilbert–Schmidt) matrix ultraproduct the available conjugation representation acts on L^2 of the ultraproduct. When that Hilbert space is infinite-dimensional, its full operator algebra is properly infinite,

so the argument no longer forces nested unitarily equivalent fixed-space projections to be equal. Section 11 makes this precise. The same distinction explains why the group $E/\langle w \rangle$ remains untouched by our argument; the ambient group G is in fact MF (Proposition 9.9). Theorem D likewise concerns only the operator-norm side: it does not bear on quasidiagonality of nuclear stably finite algebras, nor on tracial (Connes-embedding) approximation questions.

External inputs. The analytic kernel argument in Theorem A uses two non-elementary external inputs beyond elementary linear algebra and C^* -basics: property (T) of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ (Fact 2.1) and the existence and defining property of the Kazhdan projection (Fact 2.2). The advertised explicit presentation additionally uses the Conder–Robertson–Williams presentation [CRW]; the Clifford nontriviality witness is proved below by an elementary twisted-group-algebra model. Theorem C uses both quasidiagonality of amenable reduced group C^* -algebras [TWW] (Fact 2.3) and the standard quasidiagonal-to-corona embedding theorem (Fact 2.4). Theorem B uses no external input. Section 9 uses a handful of classical external inputs, each confined to the single result citing it: Borisov–Sapir residual finiteness of mapping tori of injective endomorphisms of finitely generated linear groups [BoS, Theorem 1.6] (Proposition 9.9; note that the linear case of [BoS], unlike the free case, relies on an unpublished result of Hrushovski, as its authors state); Mal’cev’s residual finiteness of finitely generated linear groups [Ma] (Corollary 9.13); B. H. Neumann’s continuum family of 2-generator groups [Neu] (Corollary 9.18); and Goryushkin’s embedding of countable groups into 2-generator simple groups [Go] (Corollary 9.14). The Haagerup–Thorbjørnsen theorem that $C_{\mathrm{red}}^*(F_k)$ is MF [HT] is cited only to populate the list of example targets in Corollary 9.4. The proof of Theorem D needs no input beyond Theorem A: the faithfulness of the canonical trace and the resulting stable finiteness are proved inline (Lemma 9.6). Its priority context uses the earlier negative solution of the general MF problem [Go23, Corollary 5.2], ultimately derived from $\mathrm{MIP}^* = \mathrm{RE}$ [JNVWY]. Remark 9.7 cites [BHV, BO] for context only.

Relation to prior work. The closest conceptual analogue is the recent work on nonsofic groups. The group-level skeleton in OpenAI’s Proposition 2.3 consists of a property-(T) subgroup Γ , one-sided containments $t_i \Gamma t_i^{-1} \leq \Gamma$, and a subgroup centralizing Γ that is moved by a compressor [OAI, Proposition 2.3]. Contemporaneous work of Kun–Thom isolates this mechanism through the compression semigroup $P_\Gamma = \{g : g\Gamma g^{-1} \leq \Gamma\}$, infranormality, normalization of centralizers, invariance of fixed algebras, and a sofic Mautner envelope [KT, Theorems A, B and C]. The property-(T) rigidity engine behind these developments — almost-invariant structure for expander sofic approximations of Kazhdan groups, upgraded and propagated — originates in Kun’s expander decomposition [Ku16] and in Kun–Thom [KT19]. Contemporaneously with [OAI, KT], and completed beforehand (see the dated footnote in

[AT]), Alekseev–Thom proved that the centralizer of any sofic embedding of a Kazhdan group is a metric ultraproduct of permutation groups, with LEF and residual-finiteness consequences [AT, Theorem A]. All of these results live on the Hamming/tracial side: they extract structural information about the approximations themselves. Lemmas 7.4, 7.5 and 7.6 below instead run the conjugation representation of operator-norm lifts and extract an exact equality of Kazhdan projections from stable finiteness; the output is not structure of an approximation but a universal marked kernel element. It also records the finite analogue: a one-sided inclusion $t\Gamma t^{-1} \leq \Gamma$ becomes an equality in every finite quotient. We therefore make no novelty claim for the one-sided-compression/lamp or Mautner-type fixed-space-propagation paradigms. In the reverse direction the divergence is equally sharp: the lamp groups of [KT] are plain abelian sums $\bigoplus_{G/\Gamma} \mathbb{Z}/2$, with no central extension, no distinguished sign, and no marked element, and the OpenAI construction uses a commuting Thompson-type witness subgroup with no Clifford structure. The marked-central-kernel device and the Clifford sign witness of the present paper are not adaptations of either.

The base group itself also has a lineage. A finitely presented Kazhdan group carrying a proper “scaling” endomorphism of an abelian normal subgroup belongs to the Cornulier–Abels lineage [Co, Abe]: Thom followed this pattern, with the shift on a central $\mathbb{F}_p[t, t^{-1}]$ - (or $\mathbb{Z}[1/p]$ -)line inside a Kazhdan matrix group, to produce hyperlinear Kazhdan groups that are not residually finite [Th, Section 2]. Our $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ with $\alpha(v, A) = (2v, A)$ is the simplest integral analogue of this scaling pattern (Remark 3.5). The approximation conclusions in [Th] are sofic or hyperlinear and use Hamming or tracial Hilbert–Schmidt models, not operator-norm coronas. Here the replacement for finite cardinality is the implication

$$P \leq \beta(t)P\beta(t)^* \implies P = \beta(t)P\beta(t)^*,$$

because the conjugation representation lands in a stably finite norm matrix corona. The central corner then converts this equality into a fixed nontrivial element killed by every corona representation.

The Clifford and marked-sign sides also have close antecedents. Slofstra uses finite and infinite-rank Clifford groups with a distinguished central involution, a shift semidirect product, and an HNN extension implementing a doubling map in an approximate-representation problem; he also explicitly records an operator-norm variant [Slo18, Section 2 and Remark 3.3]. The placement and purpose of those pieces differ here: the proper compression is in the Kazhdan base, the Clifford group supplies an external nontriviality witness, and the conclusion is a universal kernel rather than a dimension lower bound. Slofstra–Vidick’s negative-eigenspace compression for an approximate central involution is the local predecessor of Lemma 6.4 [SV, proof of Proposition 2.7]. Their Hilbert–Schmidt argument needs quantitative negative-eigenspace dimension. In the operator-norm setting, Bachner–Dogon–Lubotzky subsequently carried out the dimension-changing

corner pipeline explicitly. Their Lemmas 2.2–2.3 round an approximate involution and polar-correct an almost-unitary compression, Proposition 2.4 restricts an asymptotic representation to a nonzero almost-invariant corner, and Proposition 1.5 applies these steps to a central Deligne involution before invoking operator–Hilbert–Schmidt stability [BDL, Lemmas 2.2–2.3 and Propositions 1.5 and 2.4]. Thus changing the matrix size to the negative eigenspace, without a positive relative-rank hypothesis, is prior art. Our argument uses the same local corner technology but replaces operator–Hilbert–Schmidt stability and Mal’cev’s theorem by Kazhdan adjoint pinning and stable finiteness, producing an unconditional universal marked kernel.

There is an older analytic ancestor as well. At the foundational level, the conjugation action on Hilbert–Schmidt operators is the $\pi \otimes \bar{\pi}$ model central to Bekka’s theory of amenable unitary representations [Be]; Bekka–Valette connect this model with property (T) and the appearance of finite-dimensional subrepresentations [BV, Theorem 1]. More directly for the operator-norm approximation setting, Dadarlat’s Lemma 3.18 shows that operator-norm almost-commuting finite-rank projections force weak containment of the trivial representation in $\pi \otimes \bar{\pi}$; its proof turns the normalized projections into almost invariant normalized Hilbert–Schmidt vectors for the conjugation representation [Dad, Lemma 3.18 and its proof]. Proposition 3.19, labeled Ozawa–Thom and described there as essentially pointed out by Thom on MathOverflow, then uses property (T) to derive an infinite residually finite quotient [Dad, Proposition 3.19]. These results do not produce a universal marked kernel, but they are direct prior art for the operator-norm-to-Hilbert–Schmidt conjugation maneuver used here.

The broad exact-versus-asymptotic marked-element phenomenon is also older. Slofstra defines the quotients G^{fin} and G^{fa} by elements invisible, respectively, to finite-dimensional representations and to finite-dimensional approximate representations [Slo17, Definitions 2.5–2.7]. This is the natural precedent for the informal “MF radical” language below and for the philosophy of Theorem C. Theorem C is not a claim to originate that separation; it calibrates the same marked compression presentation used in Theorem A in the operator-norm/MF setting. Slofstra–Vidick also recall Fritz’s earlier marked BS(2, 3) hyperlinear-profile example [Fri], restated as [SV, Example 3.1], providing further Baumslag–Solitar context for the cyclic comparison.

De Chiffre–Glebsky–Lubotzky–Thom combine cohomological stability with non-residually-finite central extensions to obstruct Frobenius approximation [DGLT, Introduction]. Their norm, rigidity input, and conclusion differ from ours, but their strategy is an important predecessor for the broader architecture “designated central obstruction plus rigidity of approximate representations.” At the operator/Hilbert–Schmidt boundary, Bachner–Dogon–Lubotzky use a finite normal subgroup as a bridge between the two approximation notions: if $N \trianglelefteq \Lambda$ is finite, Λ/N is hyperlinear, and Λ is MF, then Λ is hyperlinear [BDL, Proposition 1.6]. That result does not kill N ; its proof

repairs the restriction to N and passes to a nonzero character-isotypic corner before applying Proposition 2.4 [BDL, proof of Proposition 1.6]. This is direct local prior art for the finite-normal corner. Our argument combines the corner with the one-sided Kazhdan compressor and stable finiteness to force universal annihilation of the finite subgroup, without assuming hyperlinearity of the quotient. Reformulations of (weak) soficity through equations over groups and extensions involving residually finite groups, locating potential approximation obstructions in finite (central) extensions, appear in Glebsky–Rivera and Glebsky [GR, GI]. The obstruction of Theorem A likewise lives in a finite central extension, though here the quotient $E/\langle w \rangle$ is not claimed to be residually finite.

The Kazhdan-projection compression observation needed here is proved directly in Lemmas 7.5 and 7.6; we do not attribute that observation to Carrión–Dadarlat–Eckhardt. Their work is used for the group-MF framework and for comparison examples. In particular, their treatment of Abels’ finitely presented solvable group already supplies an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18]. Since a finitely generated maximally almost periodic group is residually finite by Mal’cev’s theorem, the broad phenomenon “finitely presented amenable MF but non-MAP” is therefore not new. Theorem C instead gives a calibrated statement about the same marked-sign compression pattern used in Theorem A.

Finally, non-MF-ness of one group C^* -algebra alone does not obstruct the *group* from embedding into a corona, since group MF-ness only requires an MF quotient of $C_{\max}^*(E)$ in which the canonical image of E remains faithful. The contribution asserted here is narrower: stable-finite Kazhdan fixed-space rigidity in a norm matrix corona, together with the central-corner/marked-word device producing a specific nontrivial element in the kernel of every such representation. For context on the approximation landscape see [DGLT, BDL, OAI, KT, Sh, Dad, Slo17, Slo18, SV].

Organization. Section 2 fixes notation and states the external inputs. Section 3 constructs E . Section 4 proves $w \neq 1$ and that Γ embeds in E . Section 5 proves Theorem B. Section 6 develops the corona lemmas. Section 7 defines marked Kazhdan patterns and their compression-defect subgroups, states the finite-normal obstruction criterion (Theorem 7.2), and develops the conjugation representation; Section 8 proves the criterion and deduces Theorem A. Section 9 derives Theorem D and the further consequences. Section 10 proves Theorem C. Section 11 discusses limitations.

2. NOTATION AND EXTERNAL INPUTS

All groups are discrete and countable. All C^* -homomorphisms and C^* -algebra embeddings are understood to be unital unless explicitly stated otherwise. For a unital C^* -algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r , the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \mathrm{tr}_r(x^*x)^{1/2}$. We will use three

elementary facts without further comment: $\|x\|_2 \leq \|x\|$; $|\operatorname{tr}_r(x)| \leq \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v .

Given a sequence $(d_n)_{n \in J}$ of positive integers indexed by an infinite set $J \subseteq \mathbb{N}$, we write

$$\mathcal{Q}((d_n)_{n \in J}) = \prod_{n \in J}^{\ell^\infty} M_{d_n}(\mathbb{C}) / \bigoplus_{n \in J}^{c_0} M_{d_n}(\mathbb{C}),$$

the quotient of the bounded C^* -product by the ideal of sequences with $\|x_n\| \rightarrow 0$. We call any such algebra a *norm matrix corona*. A *corona representation* of a group E is a group homomorphism $E \rightarrow \mathcal{U}(\mathcal{Q})$ for some norm matrix corona $\mathcal{Q}$; the group E is *MF* if it admits an injective corona representation.

For a group H , $C_{\max}^*(H)$ denotes the maximal group C^* -algebra and $u_h \in C_{\max}^*(H)$ the canonical unitaries. Every unitary representation π of H on a Hilbert space extends uniquely to a $*$ -representation of $C_{\max}^*(H)$, still denoted π . Every group homomorphism $\varphi: H_1 \rightarrow H_2$ induces a unital $*$ -homomorphism $C_{\max}^*(H_1) \rightarrow C_{\max}^*(H_2)$, $u_h \mapsto u_{\varphi(h)}$.

We use the commutator convention $[x, y] = xyx^{-1}y^{-1}$.

The following two classical facts are the only non-elementary external inputs to the analytic kernel argument of Theorem A.

Fact 2.1 (property (T); [BHV, Example 1.7.4(i)]). $\operatorname{SL}_3(\mathbb{Z})$ and $\Gamma = \mathbb{Z}^3 \rtimes \operatorname{SL}_3(\mathbb{Z})$ have property (T) [BHV, Example 1.7.4(i), via Corollary 1.4.16 and Theorem 1.7.1]; an explicit-constant proof for $\operatorname{SL}_n(\mathbb{Z})$, $n \geq 3$, is [BHV, Theorem 4.2.5]. The group Γ is finitely presented — not as a consequence of property (T), which does not imply finite presentability [BHV, Section 3.4], but by the explicit presentation (3) below, built on [CRW, Theorem 2].

Fact 2.2 (Kazhdan projection; [AW, Va]). Let Γ be a group with property (T). There is a projection $p_\Gamma \in C_{\max}^*(\Gamma)$ such that for every unitary representation π of Γ on a Hilbert space $\mathcal{H}$, the operator $\pi(p_\Gamma)$ is the orthogonal projection onto the subspace $\operatorname{Fix}(\pi) = \{\xi \in \mathcal{H} : \pi(\gamma)\xi = \xi \text{ for all } \gamma \in \Gamma\}$. Moreover $u_\gamma p_\Gamma = p_\Gamma u_\gamma = p_\Gamma$ for every $\gamma \in \Gamma$.

The identities follow from the defining property: the left identity holds in every representation, and the right identity follows by taking adjoints (or replacing γ by γ^{-1}); $C_{\max}^*(\Gamma)$ has a faithful representation.

The only additional input, used exclusively in Section 10, is:

Fact 2.3 ([TWW, Corollary C]). The reduced group C^* -algebra of a countable amenable group is quasidiagonal.

Fact 2.4 (quasidiagonal embedding; [BK, Theorem 3.2.2], see also [BO, Chapter 7]). Every separable unital quasidiagonal C^* -algebra embeds unitaly into a norm matrix corona. Consequently every countable amenable group is MF.

3. THE GROUP

Throughout, $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ with multiplication $(v, A)(w, B) = (v + Aw, AB)$.

We first record concrete presentation data. By [CRW, Theorem 2], $\mathrm{SL}_3(\mathbb{Z})$ has the presentation

$$(2) \quad \left\langle x, y, z \mid \begin{array}{l} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle,$$

where the generators are realized by the matrices

$$x = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad y = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}.$$

(The matrix realization is assembled from the data of [CRW, Sections 1, 2 and 4], which set $x = h$, $y = aha$, $z = p$ in the notation there; it is not displayed as a single statement in [CRW]. We have verified directly that these three matrices satisfy the eight relations and generate $\mathrm{SL}_3(\mathbb{Z})$, which suffices since the finitely generated linear group $\mathrm{SL}_3(\mathbb{Z})$ is Hopfian [Ma].) Let $v_i = (e_i, 1)$ for $1 \leq i \leq 3$. Reading off the columns of these matrices gives the following complete finite presentation of the semidirect product:

$$(3) \quad \Gamma = \left\langle v_1, v_2, v_3, x, y, z \mid \begin{array}{l} \text{the eight relations in (2),} \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3, \quad xv_2x^{-1} = v_1, \quad xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1, \quad yv_2y^{-1} = v_2^{-1}v_3, \quad yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1}, \quad zv_2z^{-1} = v_1v_3^{-1}, \quad zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle.$$

Lemma 3.1. *The map $\alpha(v, A) = (2v, A)$ is an injective endomorphism of Γ with image $2\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ of index 8, and $a := (e_1, 1) \notin \alpha(\Gamma)$.*

Proof. We compute

$$\begin{aligned} \alpha((v, A)(w, B)) &= \alpha(v + Aw, AB) = (2v + 2Aw, AB) \\ &= (2v, A)(2w, B) = \alpha(v, A)\alpha(w, B), \end{aligned}$$

so α is a homomorphism, clearly injective, with image $2\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$; the index equals $[\mathbb{Z}^3 : 2\mathbb{Z}^3] = 8$. If $(e_1, 1) = (2v, A)$ then $A = 1$ and $e_1 = 2v$, which is impossible in $\mathbb{Z}^3$. $\square$

Let G denote the ascending HNN extension

$$(4) \quad G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \ (\gamma \in \Gamma) \rangle.$$

We will use G only as an auxiliary group receiving no injectivity claims; every statement we need about it is either a presentation identity or is verified inside an explicit linear quotient (Lemma 4.2).

Definition 3.2. Use the six base generators and the relations displayed in (3), and put $a = v_1$. Define

$$(5) \quad E = \left\langle v_1, v_2, v_3, x, y, z, t, c \left| \begin{array}{l} \text{all relations in (3),} \\ tv_it^{-1} = v_i^2 \ (1 \leq i \leq 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ c^2 = 1, \quad w^2 = 1, \\ [c, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right. \right\rangle,$$

where w abbreviates the word $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$.

Formal counterpart:

`GroupApproximation.LiteralNonMFPresentation.literal_algebraic_package.`

Thus (5) is an explicit presentation with eight generators and a fully displayed finite list of relators. The presentation identities give a canonical homomorphism $G \rightarrow E$ and a canonical (not a priori injective) homomorphism $\Gamma \rightarrow E$; we write γ, t, c for the images in E . In E the defining relations make w central and impose $w^2 = 1$; Proposition 4.3 below proves that $w \neq 1$. The relations $t\gamma t^{-1} = \alpha(\gamma)$, $[c, \gamma] = 1$ hold for all $\gamma \in \Gamma$ because they hold on generators.

Remark 3.3 (presentation robustness). Nothing in the proof of Theorem A depends on the particular presentation (3): for *any* finite presentation of Γ , together with words representing the images of the generators under α and a word representing a , the analogue of (5) satisfies Theorem A with the same proof, since the argument consumes the relations only through the canonical homomorphisms $\Gamma \rightarrow E$ and $G \rightarrow E$. The role of [CRW, Theorem 2] is therefore confined to certifying that (2) is complete as a presentation of $\text{SL}_3(\mathbb{Z})$. The two remaining ingredients of the explicit display are verified independently of [CRW]: the eight relations hold by direct integer computation, and the displayed matrices generate $\text{SL}_3(\mathbb{Z})$ because explicit words in them realize every elementary transvection — for instance $y^{-1}xzy^{-1}x^{-1}$ is the matrix $1 + E_{12}$, and the elementary matrices classically generate $\text{SL}_3(\mathbb{Z})$.

Remark 3.4 (nontriviality of the marked image forces proper infinite compression). Let ρ be any homomorphism of E to a group, and write $\Gamma' = \rho(\Gamma)$. If $\rho(t)\Gamma'\rho(t)^{-1} = \Gamma'$ — in particular whenever Γ' is finite, since a one-sided inclusion of conjugate finite subgroups is an equality — then $a' := \rho(t)^{-1}\rho(a)\rho(t) \in \Gamma'$, so

$$\rho(a(tct^{-1})a^{-1}) = \rho(t)a'\rho(c)a'^{-1}\rho(t)^{-1} = \rho(tct^{-1}),$$

because $\rho(c)$ commutes with Γ' ; hence $\rho(w) = 1$. Consequently any homomorphism with $\rho(w) \neq 1$ compresses Γ' properly, and Γ' is then infinite. This shows structurally why the pattern cannot collide with the standard

certificates of MF-ness: an infinite property-(T) image rules out amenable targets, and Theorem B rules out linear and residually finite ones.

Remark 3.5 (why this base). A base for Definition 3.2 must be a finitely presented Kazhdan group with a proper self-embedding. Infinite Kazhdan groups are typically co-Hopfian — lattices in higher-rank semisimple groups by Prasad [Pr], torsion-free hyperbolic groups by Sela [Se] — so such bases are scarce. In $\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ the abelian factor exists precisely to break co-Hopfity through the doubling endomorphism, while relative property (T) of the pair $(\Gamma, \mathbb{Z}^3)$ restores property (T) for the whole group.

4. THE WITNESS: $w \neq 1$

Clifford signs, shifts, and HNN doubling have appeared together before in approximate-representation theory, notably in Slofstra’s hyperlinear-profile construction [Slo18, Section 2]. The construction below uses the Clifford lamp only as an explicit algebraic witness: the one-sided HNN compression relevant to the obstruction remains in the separate Kazhdan base Γ .

Construction 4.1 (Clifford groups). Let X be a set. Define $\mathrm{Cl}(X)$ to be the quotient of the free unital associative real algebra on symbols $(c_x)_{x \in X}$ by the relations

$$c_x^2 = 1, \quad c_x c_y = -c_y c_x \quad (x \neq y).$$

For completeness, here is a concrete PBW model. Choose a total order on X , and let A_X be the real vector space with basis (e_F) indexed by the finite subsets $F \subseteq X$. Put

$$e_F e_G = (-1)^{\nu(F,G)} e_{F \Delta G}, \quad \nu(F,G) = \#\{(x,y) \in F \times G : y < x\}.$$

Modulo two, ν is additive in each variable with respect to symmetric difference, so this multiplication is associative and $e_\emptyset$ is the identity. The elements $e_{\{x\}}$ square to 1, and for $x \neq y$ exactly one of $\nu(\{x\}, \{y\})$ and $\nu(\{y\}, \{x\})$ is one; hence they anticommute. The defining relations therefore give a homomorphism $\mathrm{Cl}(X) \rightarrow A_X$ sending c_x to $e_{\{x\}}$. Conversely the relations reduce every word to an ordered square-free monomial, whose image is $\pm e_F$. Thus these monomials are a basis, $\mathrm{Cl}(X) \neq 0$, and $-1 \neq 1$ without invoking a normal-form theorem.

Let $\mathrm{ClLamp}(X) \leq \mathrm{Cl}(X)^\times$ be the subgroup generated by $\{\pm 1\} \cup \{c_x : x \in X\}$ and write $z = -1 \in \mathrm{ClLamp}(X)$, a central involution. Every permutation of X induces an algebra automorphism of $\mathrm{Cl}(X)$ mapping $c_x \mapsto c_{\sigma(x)}$ and fixing z ; this gives an action of any group acting on X by automorphisms of $\mathrm{ClLamp}(X)$, and a semidirect product $\mathrm{ClLamp}(X) \rtimes G$ for any G -set X . The group $\mathrm{ClLamp}(X)$ is the central sign extension of $\bigoplus_X \mathbb{Z}/2$ defined by the ordered-inversion cocycle above; its commutator pairing is $(-1)^{|F||G| - |F \cap G|}$. It is a Clifford 2-group of possibly infinite rank, familiar as the multiplicative group generated by Clifford (Majorana) generators in the CAR algebra (see, e.g., [BR2]). The same finite- and infinite-rank Clifford groups with their

distinguished central sign appear explicitly in [Slo18, Section 2]; for a modern toolbox of property (T) groups with wreath-like lamp structure see [CIOS].

Lemma 4.2 (linear model). *Embed $\Gamma \hookrightarrow \mathrm{GL}_4(\mathbb{Z})$ by $(v, A) \mapsto \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix}$ and let $D = \mathrm{diag}(2, 2, 2, 1) \in \mathrm{GL}_4(\mathbb{Z}[1/2])$. Then $D\gamma D^{-1} = \alpha(\gamma)$ for all $\gamma \in \Gamma$, so $\bar{G} := \langle \Gamma, D \rangle \leq \mathrm{GL}_4(\mathbb{Z}[1/2])$ is a quotient of G via $t \mapsto D$. In the $\bar{G}$ -set $\bar{X} := \bar{G}/\Gamma$ the cosets $D\Gamma$ and $aD\Gamma$ are distinct.*

Formal counterpart:

`GroupApproximation.ExplicitLinearModel.v1G_not_mem_range.`

Proof. $D \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} D^{-1} = \begin{pmatrix} A & 2v \\ 0 & 1 \end{pmatrix}$, which is the matrix of $\alpha(v, A)$; so the assignment $\gamma \mapsto \gamma$, $t \mapsto D$ respects the relations (4). If $aD\Gamma = D\Gamma$ then $D^{-1}aD \in \Gamma$ inside $\mathrm{GL}_4(\mathbb{Z}[1/2])$. But $D^{-1}aD = \begin{pmatrix} 1 & \frac{1}{2}e_1 \\ 0 & 1 \end{pmatrix}$ has a non-integral entry, so it does not lie in the integral group Γ . $\square$

Proposition 4.3. *$w \neq 1$ in E .*

Formal counterpart:

`GroupApproximation.LiteralNonMFLLinearWitness.literal_mark_ne_one.`

Proof. Let $\bar{X} = \bar{G}/\Gamma$ as in Lemma 4.2 and form $W := \mathrm{ClLamp}(\bar{X}) \rtimes \bar{G}$ as in Construction 4.1. Define a map on the generators of (5) by

$$\begin{aligned} g &\mapsto g \in \bar{G} \quad (g \in \{v_1, v_2, v_3, x, y, z\}), \\ t &\mapsto D \in \bar{G}, \quad c \mapsto c_o, \quad o := \Gamma \in \bar{X}. \end{aligned}$$

We check the relations. The base relations in (3) and $tgt^{-1} = \alpha(g)$ for the six base generators hold in $\bar{G}$ by Lemma 4.2. We have $c_o^2 = 1$. For $\gamma \in \Gamma$ we have $\gamma o = o$, so $\gamma c_o \gamma^{-1} = c_{\gamma o} = c_o$, giving the six commutation relations for c . Under this assignment,

$$tct^{-1} \mapsto c_{D\Gamma}, \quad a(tct^{-1})a^{-1} \mapsto c_{aD\Gamma},$$

and the two sites are distinct by Lemma 4.2, so

$$w \mapsto [c_{D\Gamma}, c_{aD\Gamma}] = c_{D\Gamma} c_{aD\Gamma} c_{D\Gamma}^{-1} c_{aD\Gamma}^{-1} = -1 = z.$$

Since z is a central involution of W , the remaining relations $w^2 = 1$ and $[w, s] = [w, t] = [w, c] = 1$ are also satisfied. Hence the assignment extends to a group homomorphism $E \rightarrow W$ sending w to $z \neq 1$, and therefore $w \neq 1$ in E . $\square$

Remark 4.4. The target uses the infinite group $\mathrm{ClLamp}(\bar{X})$ when $\bar{X}$ is infinite, but each group element involves only finitely many Clifford sites; the remaining data are matrices over $\mathbb{Z}[1/2]$. No normal-form theory for HNN extensions enters. In particular the consistency of the presentation (5) is verified by an explicit linear-algebraic model.

Corollary 4.5. *The canonical homomorphisms $\Gamma \rightarrow E$ and $\Gamma \rightarrow G$ are injective; in particular E contains an isomorphic copy of Γ .*

Proof. The homomorphism $E \rightarrow W$ constructed in the proof of Proposition 4.3 sends each of the six base generators to the corresponding element of $\bar{G} \leq W$, so the composite $\Gamma \rightarrow E \rightarrow W$ agrees on generators with, hence equals, the faithful linear embedding $\Gamma \hookrightarrow \mathrm{GL}_4(\mathbb{Z}) \leq \bar{G}$ of Lemma 4.2. A homomorphism whose composite with another homomorphism is injective is itself injective; this gives injectivity of $\Gamma \rightarrow E$, and of $\Gamma \rightarrow G$, since the latter composes with $G \rightarrow E$ to the former. $\square$

5. THE FINITE-DIMENSIONAL OBSTRUCTION

We prove Theorem B. Everything happens inside the finite-dimensional algebra $\mathrm{End}_k(V)$.

Proof of Theorem B. Let $\pi: H \rightarrow \mathrm{GL}(V)$ be a homomorphism and let

$$C = \{x \in \mathrm{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\mathrm{End}_k(V)$. Let $u = \pi(t)$ and consider the linear automorphism $\Phi(x) = uxu^{-1}$ of $\mathrm{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and set $y = u^{-1}xu$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x , and therefore

$$\pi(\gamma)y = u^{-1}\pi(t\gamma t^{-1})xu = u^{-1}x\pi(t\gamma t^{-1})u = y\pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$(6) \quad C = \Phi(C) = uCu^{-1}.$$

Now $\pi(c) \in C$ because c centralizes Γ . By (6), $\pi(tct^{-1}) = u\pi(c)u^{-1} \in C$. Since $a \in \Gamma$, the element $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi(a(tct^{-1})a^{-1}) = \pi(tct^{-1})$, and therefore

$$\pi([tct^{-1}, a(tct^{-1})a^{-1}]) = [\pi(tct^{-1}), \pi(tct^{-1})] = 1. \quad \square$$

Corollary 5.1. *The group E of Definition 3.2 admits no injective homomorphism into $\mathrm{GL}(V)$ for any finite-dimensional vector space V over any field. More strongly, finite-dimensional linear representations do not separate its points: they all identify w with 1. In particular E is not maximally almost periodic or residually finite, $C_{\max}^*(E)$ is not residually finite-dimensional as a C^* -algebra, and no finite quotient of E can witness $w \neq 1$.*

Proof. Apply Theorem B with $H = E$ and the images of Γ , t , c , a in E ; the hypotheses hold by the defining relations (5). Every finite-dimensional representation kills w , and $w \neq 1$ by Proposition 4.3. Moreover the canonical unitary u_w is not 1 in $C_{\max}^*(E)$, because the regular representation sends it to $\lambda_w \neq 1$. Every finite-dimensional $*$ -representation of $C_{\max}^*(E)$ restricts to a finite-dimensional representation of E and therefore kills $u_w - 1$; hence $C_{\max}^*(E)$ is not residually finite-dimensional. A finite quotient of E

gives a finite-dimensional representation after composition with a faithful permutation representation, so it also kills w . $\square$

Remark 5.2. The proof of Theorem B used only that a finite-dimensional subspace cannot be properly contained in an isomorphic image of itself. The corona of (1) is not finite-dimensional, and the commutant of the image of Γ there is not a coordinatewise object, so the argument does not transfer as stated. Property (T) repairs exactly this: the Kazhdan projection turns “the Γ -fixed subspace” into a single element of the corona, to which the finiteness of the corona (Lemma 6.2) can be applied. The rest of the paper carries this out.

6. NORM MATRIX CORONAS

Throughout this section $\mathcal{Q} = \mathcal{Q}((d_n)_{n \in J})$ is a norm matrix corona, $q: \prod_{n \in J} M_{d_n}(\mathbb{C}) \rightarrow \mathcal{Q}$ the quotient map.

The following lifting facts are standard perturbation theory; see Loring [Lo] for a systematic treatment.

Lemma 6.1 (lifting).

- (1) Every projection $e \in \mathcal{Q}$ lifts to a sequence $(e_n)_{n \in J}$ of projections, $e_n \in M_{d_n}(\mathbb{C})$.
- (2) Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.

Proof. (1) Lift e to a bounded sequence (x_n) ; replacing x_n by $\frac{1}{2}(x_n + x_n^*)$ and then by the functional-calculus cutoff $f(x_n)$, where $f(t) = \min(1, \max(0, t))$, we may assume $0 \leq x_n \leq 1$. These replacements preserve the quotient class because functional calculus commutes with the quotient map and $f(e) = e$. Since $q((x_n))$ is a projection, $\|x_n^2 - x_n\| \rightarrow 0$, so the spectrum of x_n is eventually contained in $[0, \frac{1}{3}] \cup (\frac{2}{3}, 1]$; for such n put $e_n = \chi_{(\frac{1}{2}, \infty)}(x_n)$, a projection with $\|e_n - x_n\| \rightarrow 0$, and set $e_n = 0$ for the finitely many remaining n . Then (e_n) lifts e .

(2) Lift u to an arbitrary bounded sequence (x_n) . Since $q((x_n))$ is unitary, $\|x_n^* x_n - 1\| \rightarrow 0$, so x_n is eventually invertible; for such n put $u_n = x_n (x_n^* x_n)^{-1/2}$, a unitary with $\|u_n - x_n\| \rightarrow 0$; set $u_n = 1$ for the remaining n . $\square$

Stable finiteness of norm matrix coronas is standard and goes back to Blackadar–Kirchberg [BK]; we include the short proofs to keep the external surface explicit.

Lemma 6.2 (stable finiteness). *Let $\mathcal{Q}$ be a norm matrix corona.*

- (1) Every isometry in $\mathcal{Q}$ is a unitary; the same holds in every matrix amplification $M_k(\mathcal{Q})$.
- (2) If $p, q \in \mathcal{Q}$ are projections with $p \leq q$ and $v^* v = q$, $vv^* = p$ for some $v \in \mathcal{Q}$, then $p = q$.

Proof. (1) Let $v \in \mathcal{Q}$ with $v^*v = 1$ and lift v to (x_n) with $\|x_n^*x_n - 1\| \rightarrow 0$. Eventually x_n is invertible (a left-invertible square matrix is invertible), and the unitary $u_n = x_n(x_n^*x_n)^{-1/2}$ satisfies $\|u_n - x_n\| \rightarrow 0$; hence $v = q((u_n))$ is unitary. For $M_k(\mathcal{Q})$ note $M_k(\mathcal{Q}) \cong \mathcal{Q}((k d_n)_{n \in J})$, since matrix amplification commutes with products and with the ideal of null sequences.

(2) This is the standard fact that in a finite algebra a projection is not equivalent to a proper subprojection; we include the short verification for completeness. As in any C^* -algebra, $v^*v = q$ gives $v = vq$ (because $\|v(1-q)\|^2 = \|(1-q)v^*v(1-q)\| = \|(1-q)q(1-q)\| = 0$) and $vv^* = p$ gives $v = pv$ (because $\|(1-p)v\|^2 = \|(1-p)vv^*(1-p)\| = 0$). From $p \leq q$ we get $qv = q(pv) = pv = v$, hence $(1-q)v = 0$ and, taking adjoints, $v^*(1-q) = 0$ and $(1-q)v^* = (v(1-q))^* = 0$. Then $s := v + (1-q)$ satisfies

$$s^*s = v^*v + v^*(1-q) + (1-q)v + (1-q) = q + 0 + 0 + (1-q) = 1,$$

so s is an isometry, and

$$ss^* = vv^* + v(1-q) + (1-q)v^* + (1-q) = p + 0 + 0 + (1-q) = p + (1-q).$$

By part (1), $ss^* = 1$, i.e. $p + (1-q) = 1$, i.e. $p = q$. $\square$

Lemma 6.3 (corner coronas). *Let $e \in \mathcal{Q}$ be a nonzero projection. Then there are an infinite set $J' \subseteq J$ and positive integers $(r_n)_{n \in J'}$ with $r_n \leq d_n$ such that $e\mathcal{Q}e$ is unitally $*$ -isomorphic to the norm matrix corona $\mathcal{Q}' = \mathcal{Q}((r_n)_{n \in J'})$, by an isomorphism carrying e to 1.*

Proof. By Lemma 6.1(1) lift e to projections (e_n) ; let $J' = \{n \in J : e_n \neq 0\}$ and $r_n = \text{rank } e_n$ for $n \in J'$. The set J' is infinite, since otherwise (e_n) would be a null sequence and $e = 0$.

Consider the map

$$\Psi: \prod_{n \in J'} e_n M_{d_n}(\mathbb{C}) e_n \longrightarrow e\mathcal{Q}e, \quad (x_n)_{n \in J'} \longmapsto q((\tilde{x}_n)_{n \in J}),$$

where $\tilde{x}_n = x_n$ for $n \in J'$ and $\tilde{x}_n = 0$ otherwise. Ψ is a $*$ -homomorphism onto $e\mathcal{Q}e$: surjectivity holds because any $y \in e\mathcal{Q}e$ lifts to (y_n) with $\|e_n y_n e_n - y_n\| \rightarrow 0$, and $e_n y_n e_n = 0$ off J' . Its kernel is exactly the sequences with $\|x_n\| \rightarrow 0$. Choosing unitary identifications $e_n \mathbb{C}^{d_n} \cong \mathbb{C}^{r_n}$ identifies $e_n M_{d_n}(\mathbb{C}) e_n$ with $M_{r_n}(\mathbb{C})$ compatibly with adjoints and norms, so

$$e\mathcal{Q}e \cong \mathcal{Q}((r_n)_{n \in J'}) = \mathcal{Q}',$$

a unital isomorphism carrying e to 1. $\square$

Lemma 6.4 (finite-normal corners). *Let H be a group, $F \trianglelefteq H$ a finite normal subgroup, and $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ a corona representation whose restriction to F is nontrivial. Then there are an infinite set $J' \subseteq J$, positive integers $(r_n)_{n \in J'}$ with $r_n \leq d_n$, and a corona representation*

$$\Theta_-: H \longrightarrow \mathcal{U}(\mathcal{Q}'), \quad \mathcal{Q}' = \mathcal{Q}((r_n)_{n \in J'}),$$

with

$$\sum_{f \in F} \Theta_-(f) = 0.$$

In the case $F = \{1, w\}$ with w a central involution and $\Theta(w) \neq 1$, the corner projection is $e = \frac{1}{2}(1 - \Theta(w))$ and the conclusion reads $\Theta_-(w) = -1$.

In the involution case, this is the norm-corona counterpart of Slofstra–Vidick’s compression to the (-1) -eigenspace of a repaired approximate central involution [SV, proof of Proposition 2.7]. In their normalized Hilbert–Schmidt setting a quantitative separation supplies a positive relative dimension. Here nonzeroness of the corona projection is enough: passing to its corner changes the ambient matrix sequence and renormalizes all later traces there. The operator-norm involution version of this corner-and-polar-correction pipeline appears in [BDL, Proposition 1.5], with rounding and compression inputs in [BDL, Lemma 2.2 and Proposition 2.4]; the character-isotypic finite-normal variant appears in the proof of [BDL, Proposition 1.6]. For general F the corner projection is the complement of the standard finite-group Reynolds projection of the finite normal subgroup.

Proof. Put

$$p_F = \frac{1}{|F|} \sum_{f \in F} \Theta(f).$$

Then $p_F^* = p_F$, because $\Theta(F)$ is closed under adjoints (F is closed under inverses), and $p_F^2 = p_F$: for each $f_0 \in F$, left translation by f_0 permutes F , so $\Theta(f_0) p_F = p_F$, and averaging over f_0 gives $p_F^2 = p_F$. Thus p_F is a projection. For $g \in H$, conjugation by g permutes the normal subgroup F , so $\Theta(g) p_F \Theta(g)^{-1} = p_F$: the projection p_F commutes with $\Theta(H)$. If $p_F = 1$ then, for every $f_0 \in F$, $\Theta(f_0) = \Theta(f_0) p_F = p_F = 1$, contradicting nontriviality of $\Theta|_F$; so $e := 1 - p_F$ is a nonzero projection commuting with $\Theta(H)$.

By Lemma 6.3, $e\mathcal{Q}e \cong \mathcal{Q}'$ by a unital $*$ -isomorphism carrying e to 1. Since e commutes with $\Theta(H)$, the map $\Theta_-(g) := \Theta(g)e$ is a unital group homomorphism $H \rightarrow \mathcal{U}(e\mathcal{Q}e) = \mathcal{U}(\mathcal{Q}')$, and

$$\sum_{f \in F} \Theta_-(f) = \left(\sum_{f \in F} \Theta(f) \right) e = |F| p_F (1 - p_F) = 0.$$

For $F = \{1, w\}$ with w central of order two, $p_F = \frac{1}{2}(1 + \Theta(w))$, so $e = \frac{1}{2}(1 - \Theta(w))$ and $\Theta_-(w) = \Theta(w)e = (1 - 2e)e = -e = -1$ in $\mathcal{Q}'$. $\square$

7. THE CONJUGATION REPRESENTATION AND THE KAZHDAN PROJECTION

The corona half of Theorem A will be deduced from a general criterion, whose data we now fix.

Definition 7.1 (marked Kazhdan pattern). Let H be a countable group. A *marked Kazhdan pattern* in H is a tuple (Λ, ι, t, c) , where Λ is a countable

group with property (T), $\iota: \Lambda \rightarrow H$ is a group homomorphism — *not* assumed injective — and $t, c \in H$, subject to

(M1) $t \iota(\lambda) t^{-1} \in \iota(\Lambda)$ for every $\lambda \in \Lambda$;

(M2) $c \iota(\lambda) = \iota(\lambda) c$ for every $\lambda \in \Lambda$.

Write $d := tct^{-1}$. The *compression-defect subgroup* of the pattern is the normal closure

$$N_{\text{comp}} = \langle\langle [d, \iota(\lambda)] : \lambda \in \Lambda \rangle\rangle_H \trianglelefteq H.$$

Formal counterpart:

`GroupApproximation.KazhdanCompressionCore.`

Formal counterpart:

`GroupApproximation.KazhdanCompressionCore.defectNormal.`

Theorem 7.2 (finite-normal obstruction criterion). *Let H be a countable group with a marked Kazhdan pattern (Λ, ι, t, c) , and let $F \trianglelefteq H$ be a finite normal subgroup with $F \subseteq N_{\text{comp}}$. Then every corona representation $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ satisfies $\Theta(f) = 1$ for all $f \in F$.*

Remark 7.3. Theorem A(2) is the instance $H = E$, $\Lambda = \Gamma$, ι the canonical homomorphism, and $F = \{1, w\}$; the membership $w \in N_{\text{comp}}$ is the identity $w = [d, \iota(a)]^2$ of Lemma 8.1 below. The property (T) hypothesis cannot simply be deleted: Theorem C realizes the pattern over the base $\mathbb{Z}$ in an MF group in which the marked involution survives. As with Theorem A, this is not a claim that every possible proof must proceed through property (T); a substitute furnishing the uniform fixed-space projection is what the argument consumes.

For the remainder of this section and in Section 8, fix a countable group H with a marked Kazhdan pattern (Λ, ι, t, c) and a finite normal subgroup $F \trianglelefteq H$ with $F \subseteq N_{\text{comp}}$, and suppose given a corona representation $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ whose restriction to F is nontrivial; the proof of Theorem 7.2 will derive a contradiction. By Lemma 6.4 there is a corona representation $\Theta_-: H \rightarrow \mathcal{U}(\mathcal{Q}')$, $\mathcal{Q}' = \mathcal{Q}((r_n)_{n \in J'})$, with $\sum_{f \in F} \Theta_-(f) = 0$. Using Lemma 6.1(2), choose for every $g \in H$ a sequence of unitaries $U_{g,n} \in M_{r_n}(\mathbb{C})$ lifting $\Theta_-(g)$, with $U_{1,n} = 1$.

Let $K_n = L^2(M_{r_n}(\mathbb{C}), \text{tr}_{r_n})$, a Hilbert space of dimension r_n^2 whose bounded operators $B(K_n)$ form the matrix algebra $M_{r_n^2}(\mathbb{C})$, and let

$$\mathcal{B} = \prod_{n \in J'} B(K_n) / \bigoplus_{n \in J'} B(K_n),$$

again a norm matrix corona. For a unitary $U \in M_{r_n}(\mathbb{C})$ let $\text{Ad } U \in \mathcal{U}(B(K_n))$ denote $x \mapsto UxU^*$.

Lemma 7.4.

(1) *For unitaries $U, V \in M_r(\mathbb{C})$ and every $x \in M_r(\mathbb{C})$,*

$$\|UxU^* - VxV^*\|_2 \leq 2 \|U - V\| \|x\|_2,$$

hence $\|\text{Ad } U - \text{Ad } V\| \leq 2\|U - V\|$ as operators on the Hilbert space K_n (and therefore as elements of $B(K_n)$).

- (2) The class $\beta(g)$ of the sequence $(\text{Ad } U_{g,n})_{n \in J'}$ in $\mathcal{B}$ defines a group homomorphism $\beta: H \rightarrow \mathcal{U}(\mathcal{B})$, independent of the choice of unitary lifts.
- (3) β extends to a unital $*$ -homomorphism $\tilde{\beta}: C_{\max}^*(H) \rightarrow \mathcal{B}$ with $\tilde{\beta}(u_g) = \beta(g)$.

Proof. (1) Since $UxU^* - VxV^* = (U - V)xU^* + Vx(U - V)^*$, the ideal property of the normalized Hilbert–Schmidt norm gives

$$\begin{aligned} \|UxU^* - VxV^*\|_2 &\leq \|(U - V)xU^*\|_2 + \|Vx(U - V)^*\|_2 \\ &\leq 2\|U - V\| \|x\|_2. \end{aligned}$$

Taking the supremum over $\|x\|_2 = 1$ proves the asserted operator-norm bound on K_n .

(2) If $(U_{g,n})$ and $(U'_{g,n})$ both lift $\Theta_-(g)$ then $\|U_{g,n} - U'_{g,n}\| \rightarrow 0$, so by (1) the two sequences of adjoint maps agree in $\mathcal{B}$; this is independence of lifts. For multiplicativity, $U_{g,n}U_{h,n}$ and $U_{gh,n}$ both lift $\Theta_-(gh)$, so $\|U_{g,n}U_{h,n} - U_{gh,n}\| \rightarrow 0$ and, by (1) again, $\beta(g)\beta(h) = \beta(gh)$. Finally $\beta(1) = 1$ since $U_{1,n} = 1$, so each $\beta(g)$ is a unitary with inverse $\beta(g^{-1})$.

(3) This is precisely the universal property of the maximal group C^* -algebra: the homomorphism $\beta: H \rightarrow \mathcal{U}(\mathcal{B})$ integrates uniquely to a unital $*$ -homomorphism $\tilde{\beta}: C_{\max}^*(H) \rightarrow \mathcal{B}$ satisfying $\tilde{\beta}(u_g) = \beta(g)$. $\square$

Let $\iota_*: C_{\max}^*(\Lambda) \rightarrow C_{\max}^*(H)$ be the unital $*$ -homomorphism induced by ι (Section 2), and let $p_\Lambda \in C_{\max}^*(\Lambda)$ be the Kazhdan projection of Fact 2.2. Composing, $\tilde{\beta} \circ \iota_*$ is a unital $*$ -homomorphism $C_{\max}^*(\Lambda) \rightarrow \mathcal{B}$, and we write

$$P := \tilde{\beta}(\iota_*(p_\Lambda)) \in \mathcal{B},$$

a projection.

Lemma 7.5 (order identity). *Let H_1 be a countable group, Λ_1 a countable group with property (T), $\iota_1: \Lambda_1 \rightarrow H_1$ a homomorphism, and $t_1 \in H_1$ an element with $t_1 \iota_1(\lambda) t_1^{-1} \in \iota_1(\Lambda_1)$ for every $\lambda \in \Lambda_1$; write $\bar{p} = (\iota_1)_*(p_{\Lambda_1}) \in C_{\max}^*(H_1)$. Then, in $C_{\max}^*(H_1)$:*

- (1) $u_{\iota_1(\lambda)} \bar{p} = \bar{p}$ for all $\lambda \in \Lambda_1$;
- (2) $\bar{p}(u_{t_1} \bar{p} u_{t_1}^*) = \bar{p}$; that is, $\bar{p} \leq u_{t_1} \bar{p} u_{t_1}^*$.

Consequently, for the standing pattern, in $\mathcal{B}$: $\beta(\iota(\lambda))P = P$ for all $\lambda \in \Lambda$, and $P \leq \beta(t)P\beta(t)^*$.

Proof. (1) is the image under $(\iota_1)_*$ of the identity $u_\lambda p_{\Lambda_1} = p_{\Lambda_1}$ of Fact 2.2, since $(\iota_1)_*(u_\lambda) = u_{\iota_1(\lambda)}$.

For (2), both sides may be evaluated in an arbitrary unitary representation π of H_1 on a Hilbert space $\mathcal{H}$, since $C_{\max}^*(H_1)$ admits a faithful one. The composition $\pi \circ (\iota_1)_*$ is the integrated form of the unitary representation $\pi \circ \iota_1$

of Λ_1 (both are $*$ -homomorphisms extending the same unitary representation), so by Fact 2.2, $\pi(\bar{p})$ is the orthogonal projection onto

$$\text{Fix}(\pi \circ \iota_1) = \{\xi \in \mathcal{H} : \pi(\iota_1(\lambda))\xi = \xi \text{ for all } \lambda \in \Lambda_1\},$$

and, since $\pi(u_{t_1}\bar{p}u_{t_1}^*) = \pi(t_1)\pi(\bar{p})\pi(t_1)^*$, the left-hand side is the orthogonal projection onto $\pi(t_1)\text{Fix}(\pi \circ \iota_1)$. We claim $\text{Fix}(\pi \circ \iota_1) \subseteq \pi(t_1)\text{Fix}(\pi \circ \iota_1)$. Let $\xi \in \text{Fix}(\pi \circ \iota_1)$ and put $\eta = \pi(t_1)^{-1}\xi$. For $\lambda \in \Lambda_1$ choose $\lambda' \in \Lambda_1$ with $t_1\iota_1(\lambda)t_1^{-1} = \iota_1(\lambda')$; then

$$\pi(\iota_1(\lambda))\eta = \pi(t_1)^{-1}\pi(t_1\iota_1(\lambda)t_1^{-1})\xi = \pi(t_1)^{-1}\pi(\iota_1(\lambda'))\xi = \pi(t_1)^{-1}\xi = \eta,$$

so $\eta \in \text{Fix}(\pi \circ \iota_1)$ and $\xi = \pi(t_1)\eta$. The inclusion of ranges gives $\pi(\bar{p}) \leq \pi(u_{t_1}\bar{p}u_{t_1}^*)$, i.e. (2) in the representation π ; as π was arbitrary and some π is faithful, (2) holds in $C_{\max}^*(H_1)$. The final statement follows by applying the $*$ -homomorphism $\tilde{\beta}$ to the instance $(H_1, \Lambda_1, \iota_1, t_1) = (H, \Lambda, \iota, t)$, for which (M1) is the required one-sided containment. $\square$

Lemma 7.6 (compression equality). $P = \beta(t)P\beta(t)^*$ in $\mathcal{B}$.

Proof. Put $Q_0 = \beta(t)P\beta(t)^*$ and $v = \beta(t)P$. Then $v^*v = P\beta(t)^*\beta(t)P = P$ and $vv^* = Q_0$, and $P \leq Q_0$ by Lemma 7.5. Apply Lemma 6.2(2) with $p := P$, $q := Q_0$, and the element v^* , which satisfies $(v^*)^*(v^*) = vv^* = Q_0$ and $(v^*)(v^*)^* = v^*v = P$; it yields $P = Q_0$. $\square$

We now pass from the corona projection P to its literal fixed-space action on a Hilbert-space ultraproduct. Fix a free ultrafilter ω on the infinite index set J' , and let

$$K_\omega = \left\{ (\xi_n)_{n \in J'} : \xi_n \in K_n, \sup_n \|\xi_n\| < \infty \right\} / \left\{ (\xi_n) : \lim_{n \rightarrow \omega} \|\xi_n\| = 0 \right\},$$

the Hilbert-space ultraproduct with inner product $\langle [\xi], [\eta] \rangle = \lim_{n \rightarrow \omega} \langle \xi_n, \eta_n \rangle$.

Lemma 7.7. *For a bounded sequence $(T_n)_{n \in J'}$ of operators with class $[T] \in \mathcal{B}$, the assignment $\sigma([T])([\xi_n]) = [(T_n\xi_n)]$ is a well-defined unital $*$ -homomorphism $\sigma : \mathcal{B} \rightarrow B(K_\omega)$. Moreover $\sigma(P)$ is the orthogonal projection onto*

$$\text{Fix}_\omega = \{\zeta \in K_\omega : \sigma(\beta(\iota(\lambda)))\zeta = \zeta \text{ for all } \lambda \in \Lambda\}.$$

Proof. If (T_n) is a null sequence in operator norm then $\|T_n\xi_n\| \leq \|T_n\|\sup\|\xi_n\| \rightarrow 0$, in particular along ω ; so σ is well defined on $\mathcal{B}$, and it is clearly a unital $*$ -homomorphism. The composition $\sigma \circ \tilde{\beta} \circ \iota_*$, from $C_{\max}^*(\Lambda)$ to $B(K_\omega)$, is a $*$ -representation on a Hilbert space extending the unitary representation $\lambda \mapsto \sigma(\beta(\iota(\lambda)))$; by uniqueness of integrated forms and Fact 2.2,

$$\sigma(P) = \sigma(\tilde{\beta}(\iota_*(p_\Lambda)))$$

is the orthogonal projection onto the fixed space of that representation, which is Fix_ω . $\square$

8. PROOF OF THE CRITERION AND OF THEOREM A

Proof of Theorem 7.2. Suppose, for contradiction, that some corona representation $\Theta: H \rightarrow \mathcal{U}(\mathcal{Q})$ is nontrivial on F , and adopt all notation fixed in Section 7: the corner representation Θ_- with $\sum_{f \in F} \Theta_-(f) = 0$, the unitary lifts $U_{g,n}$, the corona $\mathcal{B}$, the homomorphism β , the projection P , the ultraproduct K_ω and the representation σ .

Step 1: the lamp vector is fixed. Each $U_{c,n}$ is a unitary matrix, hence a vector in K_n of norm $\|U_{c,n}\|_2 = 1$; let $\xi_c = [(U_{c,n})] \in K_\omega$, a unit vector. For $\lambda \in \Lambda$ we have $\iota(\lambda) c \iota(\lambda)^{-1} = c$ in H by (M2), so $U_{\iota(\lambda),n} U_{c,n} U_{\iota(\lambda),n}^*$ and $U_{c,n}$ both lift $\Theta_-(c)$, whence $\|U_{\iota(\lambda),n} U_{c,n} U_{\iota(\lambda),n}^* - U_{c,n}\| \rightarrow 0$ and a fortiori the K_n -norms tend to 0. Therefore $\sigma(\beta(\iota(\lambda)))\xi_c = \xi_c$ for every $\lambda \in \Lambda$, i.e. $\xi_c \in \text{Fix}_\omega$, and by Lemma 7.7, $\sigma(P)\xi_c = \xi_c$.

Step 2: the translated lamp vector is fixed. Let $\xi_d := \sigma(\beta(t))\xi_c = [(D_n)]$, where $D_n := U_{t,n} U_{c,n} U_{t,n}^*$, an operator-norm lift of $\Theta_-(d)$. By Lemma 7.6 and Step 1,

$$\sigma(P)\xi_d = \sigma(\beta(t)P\beta(t)^*)\sigma(\beta(t))\xi_c = \sigma(\beta(t))\sigma(P)\xi_c = \xi_d,$$

so $\xi_d \in \text{Fix}_\omega$: for every $\lambda \in \Lambda$,

$$(7) \quad \lim_{n \rightarrow \omega} \|U_{\iota(\lambda),n} D_n U_{\iota(\lambda),n}^* - D_n\|_2 = 0.$$

Step 3: every defect commutator collapses in Hilbert–Schmidt norm. For $\lambda \in \Lambda$ put

$$X_{\lambda,n} := D_n U_{\iota(\lambda),n} D_n^{-1} U_{\iota(\lambda),n}^{-1},$$

a unitary in $M_{r_n}(\mathbb{C})$ and an operator-norm lift of $\Theta_-([d, \iota(\lambda)])$. Two applications of unitary invariance of $\|\cdot\|_2$ give

$$\|X_{\lambda,n} - 1\|_2 = \|D_n U_{\iota(\lambda),n} - U_{\iota(\lambda),n} D_n\|_2 = \|D_n - U_{\iota(\lambda),n} D_n U_{\iota(\lambda),n}^{-1}\|_2,$$

so by (7),

$$(8) \quad \lim_{n \rightarrow \omega} \|X_{\lambda,n} - 1\|_2 = 0 \quad \text{for every } \lambda \in \Lambda.$$

Step 4: the collapse propagates to the compression-defect subgroup. Let $f \in F$. Since $F \subseteq N_{\text{comp}}$, there are finitely many (possibly zero) elements $h_1, \dots, h_L \in H$, $\lambda_1, \dots, \lambda_L \in \Lambda$ and signs $\varepsilon_1, \dots, \varepsilon_L \in \{\pm 1\}$ with

$$f = \prod_{i=1}^L h_i [d, \iota(\lambda_i)]^{\varepsilon_i} h_i^{-1}.$$

Then

$$V_{f,n} := \prod_{i=1}^L U_{h_i,n} X_{\lambda_i,n}^{\varepsilon_i} U_{h_i,n}^*$$

is an operator-norm lift of $\Theta_-(f)$, and the telescoping estimate $\|AB - 1\|_2 \leq \|A - 1\|_2 + \|B - 1\|_2$ for unitaries A, B , together with unitary invariance ($\|UXU^* - 1\|_2 = \|X - 1\|_2$ and $\|X^{-1} - 1\|_2 = \|X - 1\|_2$), gives

$$\|V_{f,n} - 1\|_2 \leq \sum_{i=1}^L \|X_{\lambda_i,n} - 1\|_2 \xrightarrow{n \rightarrow \omega} 0$$

by (8).

Step 5: the corner average forbids the collapse. Each $V_{f,n}$ lifts $\Theta_-(f)$, so the finite sum $\sum_{f \in F} V_{f,n}$ is an operator-norm lift of $\sum_{f \in F} \Theta_-(f) = 0$:

$$\left\| \sum_{f \in F} V_{f,n} \right\| \longrightarrow 0 \quad (n \rightarrow \infty \text{ in } J'),$$

hence also along ω . On the other hand, by Step 4,

$$\left\| \sum_{f \in F} V_{f,n} - |F| \right\|_2 \leq \sum_{f \in F} \|V_{f,n} - 1\|_2 \xrightarrow{n \rightarrow \omega} 0,$$

so, using $\|x\|_2 \leq \|x\|$,

$$|F| = \left\| |F| \cdot 1 \right\|_2 \leq \left\| \sum_{f \in F} V_{f,n} - |F| \right\|_2 + \left\| \sum_{f \in F} V_{f,n} \right\| \xrightarrow{n \rightarrow \omega} 0,$$

which is absurd since $|F| \geq 1$. This contradiction proves Theorem 7.2. $\square$

Lemma 8.1 (the marked word is a defect square). *Let H_1 be a group and $d, a_1 \in H_1$ with $d^2 = 1$. Then*

$$[d, a_1 d a_1^{-1}] = [d, a_1]^2.$$

Formal counterpart:

`GroupApproximation.MarkedCompressionInclusionData.word_eq_compressionDefect_sq.`

Proof. Put $b = a_1 d a_1^{-1}$, so $b^2 = a_1 d^2 a_1^{-1} = 1$; thus $b^{-1} = b$ and $d^{-1} = d$. Then $[d, a_1] = d a_1 d^{-1} a_1^{-1} = d (a_1 d a_1^{-1}) = db$, and

$$[d, a_1]^2 = (db)^2 = db db = d b d^{-1} b^{-1} = [d, b]. \quad \square$$

Proof of Theorem A. Part (1) is Proposition 4.3. For part (2), let $\iota: \Gamma \rightarrow E$ be the canonical homomorphism and $a = v_1 = (e_1, 1)$. The tuple (Γ, ι, t, c) is a marked Kazhdan pattern in E : Γ has property (T) by Fact 2.1; (M1) holds because $t \iota(\gamma) t^{-1} = \iota(\alpha(\gamma))$ for every $\gamma \in \Gamma$ (both sides are homomorphisms in γ , and (5) gives the identity on the six base generators); (M2) holds because $[c, g] = 1$ in (5) for the six base generators g . In E we have $d^2 = (t c t^{-1})^2 = t c^2 t^{-1} = 1$, so Lemma 8.1 with $a_1 = \iota(a)$ gives

$$w = [d, \iota(a) d \iota(a)^{-1}] = [d, \iota(a)]^2 \in N_{\text{comp}},$$

since $[d, \iota(a)]$ is one of the defining generators of N_{comp} . The subgroup $F = \{1, w\} = \langle w \rangle$ is finite and normal: centrality and $w^2 = 1$ are defining relations, while Proposition 4.3 gives $w \neq 1$. Moreover $F \subseteq N_{\text{comp}}$. Theorem 7.2 therefore gives $\Theta(w) = 1$ for every corona representation Θ of E , which is part (2).

For the final assertions of Theorem A: an MF embedding of E would be an injective corona representation, which is impossible because every corona representation sends the nontrivial element w to 1. If $C_{\text{red}}^*(E)$ were an MF algebra, a unital C^* -embedding $C_{\text{red}}^*(E) \hookrightarrow \mathcal{Q}$ composed with the canonical map $E \rightarrow \mathcal{U}(C_{\text{red}}^*(E))$, would exhibit an injective corona representation of E : the map $g \mapsto \lambda_g$ is injective, although the quotient map $C_{\text{max}}^*(E) \rightarrow C_{\text{red}}^*(E)$ need not be faithful. The canonical map $E \rightarrow \mathcal{U}(C_{\text{max}}^*(E))$ is also injective, since its composition with the regular representation is $g \mapsto \lambda_g$; this treats $C_{\text{max}}^*(E)$. $\square$

Remark 8.2. The proof shows more than non-injectivity of each individual Θ : the fixed element w lies in the kernel of *every* corona representation of E , so every corona representation factors through $E/\langle w \rangle$ (note $\langle w \rangle = \{1, w\}$ is central, hence normal). In the language of approximation radicals, w is a nontrivial element of the “MF radical” of E . This terminology is meant in analogy with Slofstra’s earlier quotients G^{fin} and G^{fa} , obtained by killing elements invisible to exact and approximate finite-dimensional representations, respectively [Slo17, Definitions 2.5–2.7]. By Theorem 7.2 the same holds in every countable group with a marked Kazhdan pattern: every finite normal subgroup of the compression-defect subgroup lies in the MF radical. Section 9 develops the radical systematically; we do not know whether $E/\langle w \rangle$ is MF, i.e. whether the MF radical of E is exactly $\{1, w\}$.

Remark 8.3. No lower bound on the corner ranks r_n , or on their density in d_n , is used anywhere: Lemma 6.4 renormalizes the trace so that all subsequent estimates are relative to the corner. This is what makes the operator-norm setting insensitive to small relative ranks, which block the same argument in Hilbert–Schmidt metrics (Section 11).

9. CONSEQUENCES

Throughout this section, “corona representation” and “MF” are as in Section 2.

9.1. The MF radical.

Definition 9.1. For a countable group G_1 , the *MF radical* is

$$\text{Rad}_{\text{MF}}(G_1) = \bigcap \{ \ker \Theta : \Theta \text{ a corona representation of } G_1 \},$$

a normal subgroup of G_1 . This is the operator-norm analogue of the invisible subgroups of [Slo17, Definitions 2.5–2.7] and of the approximation kernels studied in [NST].

Lemma 9.2 (portability). *For every homomorphism $\varphi: G_1 \rightarrow G_2$ of countable groups,*

$$\varphi(\text{Rad}_{\text{MF}}(G_1)) \subseteq \text{Rad}_{\text{MF}}(G_2):$$

the MF radical is functorial. In particular, if $\varphi: E \rightarrow G_2$ satisfies $\varphi(w) \neq 1$, then G_2 is not MF.

Proof. If $x \in \text{Rad}_{\text{MF}}(G_1)$ and Θ is a corona representation of G_2 , then $\Theta \circ \varphi$ is a corona representation of G_1 , so $\Theta(\varphi(x)) = 1$. For the second statement: $w \in \text{Rad}_{\text{MF}}(E)$ by Theorem A, so $1 \neq \varphi(w) \in \text{Rad}_{\text{MF}}(G_2)$, and an injective corona representation of G_2 would kill $\varphi(w)$, a contradiction. $\square$

Proposition 9.3 (the largest MF quotient). *Let G_1 be a countable group and $R = \text{Rad}_{\text{MF}}(G_1)$.*

(1) *R is the kernel of a single corona representation of G_1 .*

(2) *G_1/R is MF.*

(3) *Every homomorphism from G_1 to an MF group factors through G_1/R .*

In particular G_1 is MF if and only if $R = \{1\}$. The statement is presumably folklore once the radical is defined; we include a proof for completeness and claim no novelty for it.

Proof. (3) If $\varphi: G_1 \rightarrow M$ with M MF and $j: M \rightarrow \mathcal{U}(\mathcal{Q})$ is an injective corona representation, then $j \circ \varphi$ is a corona representation of G_1 , so $j(\varphi(R)) = 1$ and, by injectivity of j , $\varphi(R) = 1$.

(1) and (2). If $R = G_1$, the trivial representation has kernel R and G_1/R is trivial, hence MF; so assume $R \neq G_1$ and let $(g_i)_{i \geq 1}$ list, possibly with repetitions, elements of $G_1 \setminus R$ meeting every nontrivial coset of R . For each i choose a corona representation $\Theta_i: G_1 \rightarrow \mathcal{U}(\mathcal{Q}_i)$, $\mathcal{Q}_i = \mathcal{Q}((d_n^{(i)})_{n \in J_i})$, with $\Theta_i(g_i) \neq 1$; put $\varepsilon_i = \frac{1}{2} \|\Theta_i(g_i) - 1\| > 0$, and choose unitary lifts $(U_{g,n}^{(i)})_{n \in J_i}$ of $\Theta_i(g)$ for every $g \in G_1$ (Lemma 6.1), with $U_{1,n}^{(i)} = 1$. Recall that the quotient norm in a corona is the limit superior of the coordinate norms: subtracting a null sequence does not change the limit superior, and truncating finitely many coordinates realizes it.

Enumerate $G_1 = \{h_1, h_2, \dots\}$ and fix a stage $k \geq 1$. For each $i \leq k$: the finitely many products $U_{g,n}^{(i)} U_{h,n}^{(i)}$ and $U_{gh,n}^{(i)}$ with $g, h \in \{h_1, \dots, h_k\}$ lift the same corona elements, so there is n_0 with all the corresponding defects at most $1/k$ for $n \geq n_0$; and since $\limsup_{n \in J_i} \|U_{g_i,n}^{(i)} - 1\| = 2\varepsilon_i$, there are infinitely many $n \in J_i$ with $\|U_{g_i,n}^{(i)} - 1\| \geq \varepsilon_i$. Choose $n_i(k) \in J_i$ satisfying both conditions and set

$$\Phi_k(g) = \bigoplus_{i \leq k} U_{g, n_i(k)}^{(i)} \in \mathcal{U}(M_{D_k}(\mathbb{C})), \quad D_k = \sum_{i \leq k} d_{n_i(k)}^{(i)}.$$

Define $\Theta_\infty(g)$ as the class of $(\Phi_k(g))_{k \in \mathbb{N}}$ in $\mathcal{Q}((D_k)_{k \in \mathbb{N}})$. For fixed $g, h \in G_1$, both lie in $\{h_1, \dots, h_k\}$ for all large k , and the operator norm of a block sum is the maximum of the block norms, so $\|\Phi_k(g)\Phi_k(h) - \Phi_k(gh)\| \leq 1/k \rightarrow 0$;

together with $\Phi_k(1) = 1$ this makes Θ_∞ a corona representation of G_1 . Consequently $\Theta_\infty(R) = 1$. Conversely, if $g \notin R$, pick i with $g \in g_i R$; then $\Theta_\infty(g) = \Theta_\infty(g_i)$, and for every $k \geq i$, $\|\Phi_k(g_i) - 1\| \geq \|U_{g_i, n_i(k)}^{(i)} - 1\| \geq \varepsilon_i$, so $\|\Theta_\infty(g_i) - 1\| \geq \varepsilon_i$ and $\Theta_\infty(g) \neq 1$. Hence $\ker \Theta_\infty = R$, proving (1), and Θ_∞ descends to an injective corona representation of G_1/R , proving (2). For the final equivalence: if G_1 is MF, an injective corona representation forces $R = \{1\}$; if $R = \{1\}$, then (2) says $G_1 = G_1/R$ is MF. $\square$

9.2. Targets beyond coronas.

Corollary 9.4 (subalgebras of coronas). *Let $H, (\Lambda, \iota, t, c)$ and $F \subseteq N_{\text{comp}}$ be as in Theorem 7.2. Let A be a nonzero C^* -subalgebra of a norm matrix corona $\mathcal{Q}$ whose unit $p := 1_A$ is not necessarily the unit of $\mathcal{Q}$, and let $\Theta: H \rightarrow \mathcal{U}(A)$ be a group homomorphism. Then $\Theta(f) = 1_A$ for every $f \in F$. The same conclusion holds when A is a unital separable MF algebra — for instance a unital separable quasidiagonal algebra (Fact 2.4), in particular an AF algebra, or $C_{\text{red}}^*(F_k)$ for a free group F_k [HT, Corollary 8.4].*

Proof. Every unitary u of A satisfies $u = pup$ and $u^*u = uu^* = p$, so $\mathcal{U}(A) \subseteq \mathcal{U}(p\mathcal{Q}p)$, and $p \neq 0$. By Lemma 6.3 there is a unital $*$ -isomorphism of $p\mathcal{Q}p$ onto a norm matrix corona carrying p to the unit. Composing, Θ becomes a corona representation of H , and Theorem 7.2 gives $\Theta(f) = 1_A$ for $f \in F$. If A is a unital separable MF algebra, fix a $*$ -monomorphism $\psi: A \rightarrow \mathcal{Q}_1$ into a norm matrix corona, unital or not [BK]; then $\psi(A)$ is a C^* -subalgebra of $\mathcal{Q}_1$ with unit $\psi(1_A)$, and applying the first part to $\psi \circ \Theta$ gives $\psi(\Theta(f)) = \psi(1_A)$, whence $\Theta(f) = 1_A$ by injectivity. Unital separable quasidiagonal algebras embed unitaly into coronas by Fact 2.4, AF algebras are quasidiagonal, and $C_{\text{red}}^*(F_k)$ is MF for every $k \in \mathbb{N} \cup \{\infty\}$ by [HT, Corollary 8.4]. $\square$

Corollary 9.5. *The group E admits no injective homomorphism into the unitary group of any C^* -subalgebra of any norm matrix corona — in particular, none into $\mathcal{U}(A)$ for A a unital separable MF algebra, a unital separable quasidiagonal algebra, or $C_{\text{red}}^*(F_k)$.*

Proof. By the proof of Theorem A, the pattern (Γ, ι, t, c) and $F = \{1, w\}$ satisfy the hypotheses of Corollary 9.4 in $H = E$; hence any such homomorphism kills w , while $w \neq 1$ by Proposition 4.3. $\square$

9.3. Stable finiteness does not imply MF.

Lemma 9.6. *Let G_1 be a countable discrete group.*

- (1) *The canonical tracial state τ on $C_{\text{red}}^*(G_1)$ is faithful.*
- (2) *Every unital C^* -algebra A with a faithful tracial state is stably finite: for every $k \geq 1$, every isometry in $M_k(A)$ is a unitary.*

Proof. (1) Realize $C_{\text{red}}^*(G_1) \subseteq B(\ell^2 G_1)$ as the norm closure of the span of the left translation unitaries λ_g , and let ρ_g be the right translation unitaries,

$(\rho_g \xi)(h) = \xi(hg)$; each ρ_g commutes with each $\lambda_{g'}$, hence with every element of $C_{\text{red}}^*(G_1)$. The canonical trace is $\tau(x) = \langle x\delta_e, \delta_e \rangle$. If $\tau(x^*x) = 0$ then $\|x\delta_e\| = 0$, so $x\delta_e = 0$, and then $x\delta_g = x\rho_{g^{-1}}\delta_e = \rho_{g^{-1}}x\delta_e = 0$ for every $g \in G_1$. The vectors δ_g are total in $\ell^2 G_1$, so $x = 0$.

(2) On $M_k(A)$ the functional $\tau_k([x_{ij}]) = \frac{1}{k} \sum_i \tau(x_{ii})$ is a state with the trace property $\tau_k(xy) = \frac{1}{k} \sum_{i,j} \tau(x_{ij}y_{ji}) = \tau_k(yx)$, and it is faithful: the i -th diagonal entry of x^*x is $\sum_j x_{ji}^*x_{ji}$, so $\tau_k(x^*x) = 0$ forces $\tau(x_{ji}^*x_{ji}) = 0$, hence $x_{ji} = 0$, for all i, j . If $v \in M_k(A)$ satisfies $v^*v = 1$, then $y := 1 - vv^*$ is a projection with $\tau_k(y) = \tau_k(1) - \tau_k(vv^*) = \tau_k(1) - \tau_k(v^*v) = 0$; since $y = y^*y$, faithfulness gives $y = 0$, i.e. $vv^* = 1$. $\square$

Proof of Theorem D. $C_{\text{red}}^*(E)$ is unital and separable since E is countable; by Lemma 9.6 it carries a faithful tracial state and is stably finite. It is not an MF algebra by Theorem A. $\square$

Remark 9.7. Blackadar and Kirchberg proved that MF algebras are stably finite [BK]. The general converse was already refuted as a consequence of the negative solution to the Connes embedding problem [Go23, Corollary 5.2]; Theorem D instead realizes that failure as the reduced group C^* -algebra of an explicit finitely presented group, with a faithful trace. The example is not nuclear: Γ embeds into E (Corollary 4.5); the infinite group Γ has property (T) and is therefore not amenable, since amenable groups with property (T) are compact, hence finite when discrete [BHV, Theorem 1.1.6]; and the reduced C^* -algebra of a nonamenable group is not nuclear [BO, Theorem 2.6.8]. The nuclear territory is untouched: whether every stably finite separable *nuclear* C^* -algebra is quasidiagonal remains open, and by [TWW] every faithful trace on a separable nuclear C^* -algebra satisfying the UCT is quasidiagonal.

9.4. The ambient HNN extension.

Lemma 9.8 (residually finite groups are MF). *Every countable residually finite group is MF.*

Proof. Let G_1 be countable and residually finite; if G_1 is trivial the statement is trivially true, so enumerate its nontrivial elements $x_1, x_2, \dots$ (with repetitions if G_1 is finite). For each k choose a finite-index normal subgroup $N_k \trianglelefteq G_1$ with $x_k \notin N_k$, put $M_k = N_1 \cap \dots \cap N_k$, a finite-index normal subgroup, and let $\varphi_k: G_1 \rightarrow G_1/M_k$ be the quotient map onto the finite group G_1/M_k . Let π_k be the left regular representation of G_1/M_k composed with φ_k , a finite-dimensional unitary representation of G_1 . Define $\Theta(g)$ as the class of $(\pi_k(g))_k$ in $\mathcal{Q}((|G_1/M_k|)_k)$; each π_k is an exact representation, so Θ is a corona representation. For $g = x_i$ and $k \geq i$ the element $\varphi_k(x_i)$ is nontrivial, so the permutation matrix $\pi_k(x_i)$ moves some standard basis vector δ , giving $\|(\pi_k(x_i) - 1)\delta\| = \sqrt{2}$ and $\|\pi_k(x_i) - 1\| \geq \sqrt{2}$. Hence $\|\Theta(x_i) - 1\| \geq \sqrt{2}$ and Θ is injective. $\square$

Proposition 9.9 (the ambient group). *Let G be the ascending HNN extension (4).*

- (1) $C_{\max}^*(G)$ contains an explicitly described nonunitary isometry. In particular $C_{\max}^*(G)$ is not finite, a fortiori not stably finite, and not an MF algebra.
- (2) G is residually finite, hence MF.

Proof. (1) Write $\iota_G: \Gamma \rightarrow G$ for the canonical homomorphism and $\bar{p} = (\iota_G)_*(p_\Gamma) \in C_{\max}^*(G)$ for the image of the Kazhdan projection. The data (G, Γ, ι_G, t) satisfy the hypotheses of Lemma 7.5, because $t\iota_G(\gamma)t^{-1} = \iota_G(\alpha(\gamma))$; hence $\bar{p} \leq Q_1 := u_t \bar{p} u_t^*$. Put

$$s = \bar{p} u_t^* + (1 - Q_1).$$

From $\bar{p} \leq Q_1$ we get $\bar{p}Q_1 = Q_1\bar{p} = \bar{p}$, hence $(1 - Q_1)\bar{p} = 0$ and $\bar{p}(1 - Q_1) = 0$; also $Q_1 u_t \bar{p} = u_t \bar{p} u_t^* u_t \bar{p} = u_t \bar{p}$, so $(1 - Q_1)u_t \bar{p} = 0$ and, taking adjoints, $\bar{p} u_t^* (1 - Q_1) = 0$. Therefore

$$\begin{aligned} s^* s &= u_t \bar{p} \bar{p} u_t^* + u_t \bar{p} (1 - Q_1) + (1 - Q_1) \bar{p} u_t^* + (1 - Q_1) \\ &= Q_1 + (1 - Q_1) = 1 \end{aligned}$$

and

$$\begin{aligned} ss^* &= \bar{p} u_t^* u_t \bar{p} + \bar{p} u_t^* (1 - Q_1) + (1 - Q_1) u_t \bar{p} + (1 - Q_1) \\ &= \bar{p} + (1 - Q_1) = 1 - (Q_1 - \bar{p}) : \end{aligned}$$

s is an isometry with defect $1 - ss^* = Q_1 - \bar{p}$.

It remains to check $Q_1 \neq \bar{p}$. Let $H_0 = t\iota_G(\Gamma)t^{-1} \leq G$ and let π be the quasi-regular representation of G on $\ell^2(G/H_0)$, with basepoint $b = H_0 \in G/H_0$. As in the proof of Lemma 7.5, $\pi(\bar{p})$ is the orthogonal projection onto the $\iota_G(\Gamma)$ -invariant vectors, and $\pi(Q_1) = \pi(t)\pi(\bar{p})\pi(t)^*$. The vector $\pi(t)^{-1}\delta_b = \delta_{t^{-1}H_0}$ is $\iota_G(\Gamma)$ -invariant: for $\gamma \in \Gamma$,

$$\iota_G(\gamma)t^{-1}H_0 = t^{-1}(t\iota_G(\gamma)t^{-1})H_0 = t^{-1}H_0,$$

since $t\iota_G(\gamma)t^{-1} \in H_0$; hence $\pi(Q_1)\delta_b = \delta_b$. On the other hand $\pi(\iota_G(a))\delta_b = \delta_{aH_0} \neq \delta_b$: if $aH_0 = H_0$ then $a \in H_0$, and applying the homomorphism $G \rightarrow \bar{G}$ of Lemma 4.2, which restricts to the identity on Γ and sends t to D , would give $a \in D\Gamma D^{-1} = \alpha(\Gamma)$ inside $\mathrm{GL}_4(\mathbb{Z}[1/2])$, contradicting Lemma 3.1. So δ_b is not $\iota_G(\Gamma)$ -invariant, i.e. $\pi(\bar{p})\delta_b \neq \delta_b$ while $\pi(Q_1)\delta_b = \delta_b$; hence $Q_1 \neq \bar{p}$ and s is not a unitary.

If $C_{\max}^*(G)$ embedded unitally into a norm matrix corona, the image of s would be an isometry there, hence a unitary by Lemma 6.2(1), and injectivity would force $ss^* = 1$, a contradiction; so $C_{\max}^*(G)$ is not an MF algebra, and the isometry s itself witnesses the failure of finiteness.

(2) G is the mapping torus of the injective endomorphism α of the finitely generated linear group $\Gamma \leq \mathrm{GL}_4(\mathbb{Z})$, so G is residually finite by Borisov–Sapir [BoS, Theorem 1.6]; MF-ness follows from Lemma 9.8. $\square$

Remark 9.10. Proposition 9.9 separates group-level and C^* -level MF-ness in the direction opposite to Theorem D: G is an MF group whose maximal

group C^* -algebra is not even finite, while E is a non-MF group whose reduced group C^* -algebra is stably finite. The one-way implication — if $C_{\max}^*$ or C_{red}^* of a group is an MF algebra then the group is MF — is recorded in [Sh].

9.5. Failure of permanence properties.

Corollary 9.11 (quotients). *MF groups are not closed under quotients: the free group F_8 is MF, and E is a quotient of F_8 .*

Proof. E is generated by eight elements (5), hence is a quotient of F_8 . Free groups are residually finite: F_8 embeds into F_2 (a subgroup of index 7 of F_2 is free of rank 8 by the classical Nielsen–Schreier theorem), F_2 embeds into $\text{SL}_2(\mathbb{Z})$ by Sanov’s classical ping-pong pair $(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}), (\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix})$, residual finiteness passes to subgroups, and $\text{GL}_2(\mathbb{Z})$ is residually finite by reduction modulo primes (a nontrivial integer matrix stays nontrivial modulo any prime exceeding the absolute values of the entries of its difference from the identity). So F_8 is MF by Lemma 9.8, while E is not MF by Theorem A. $\square$

Proposition 9.12 (the witness group is non-MF). *Let $W = \text{CLamp}(\bar{X}) \rtimes \bar{G}$ be the witness group of Proposition 4.3. Then W is finitely generated, the homomorphism $E \rightarrow W$ of that proof is surjective, and the central sign $z = -1 \in \text{CLamp}(\bar{X})$ satisfies $z \in \text{Rad}_{\text{MF}}(W) \setminus \{1\}$. In particular W is not MF.*

Proof. The image of the homomorphism $E \rightarrow W$ contains the six base generators of Γ , hence Γ ; it contains D ; so it contains $\bar{G} = \langle \Gamma, D \rangle$. It contains c_o , hence $c_g \Gamma = g c_o g^{-1}$ for every $g \in \bar{G}$, which is all of $\{c_x : x \in \bar{X}\}$ since $\bar{G}$ acts transitively on $\bar{X} = \bar{G}/\Gamma$. Finally $(c_{D\Gamma} c_{aD\Gamma})^2 = -c_{D\Gamma}^2 c_{aD\Gamma}^2 = -1 = z$ because the two sites are distinct (Lemma 4.2), so the image contains z and therefore all of $\text{CLamp}(\bar{X}) = \langle \pm 1, c_x : x \in \bar{X} \rangle$: the map is surjective, and W is generated by the images of the eight generators of E . The proof of Proposition 4.3 sends w to z , so Lemma 9.2 gives $z \in \text{Rad}_{\text{MF}}(W)$, and $z \neq 1$ in W by Construction 4.1. A group whose MF radical is nontrivial admits no injective corona representation. $\square$

Corollary 9.13 (extensions). *MF groups are not closed under extensions, even split ones: in the split exact sequence*

$$1 \longrightarrow \text{CLamp}(\bar{X}) \longrightarrow W \longrightarrow \bar{G} \longrightarrow 1,$$

the kernel is locally finite, hence amenable and MF; the quotient $\bar{G} \leq \text{GL}_4(\mathbb{Z}[1/2])$ is a finitely generated linear group, hence residually finite by Mal’cev’s theorem [Ma] and MF by Lemma 9.8; and W is not MF (Proposition 9.12). In particular there is a finitely generated non-MF group that is (locally finite)-by-(residually finite). This contrasts with the positive permanence theory for MF groups in [Sh].

Proof. Exactness and splitting hold by construction of the semidirect product. Every finite subset of $\text{CLamp}(\bar{X})$ lies in the subgroup generated by ± 1 and finitely many c_x , a finite 2-group, as in the proof of Theorem C; so

$\text{Cilamp}(\bar{X})$ is locally finite, hence amenable, hence MF by Facts 2.3 and 2.4. The remaining assertions were proved above. $\square$

Corollary 9.14 (a simple envelope). *There exists a 2-generator simple group S with $\text{Rad}_{\text{MF}}(S) = S$: every corona representation of S , and every homomorphism from S into any MF group, is trivial.*

Proof. By Goryushkin's theorem [Go], the countable group E embeds into a 2-generator simple group S . By Lemma 9.2 applied to the embedding, the image of w is a nontrivial element of $\text{Rad}_{\text{MF}}(S)$. The radical is a normal subgroup of the simple group S , so $\text{Rad}_{\text{MF}}(S) = S$, i.e. every corona representation of S is trivial. If $\varphi: S \rightarrow M$ is a homomorphism to an MF group and j an injective corona representation of M , then $j \circ \varphi$ is a corona representation, hence trivial, so φ is trivial. The simple-envelope theorem is entirely Goryushkin's; what is added here is the nontrivial portable radical element. $\square$

9.6. The scaling family. For an integer $m \geq 2$ let E_m be the group with the presentation (5), except that the three relations $tv_it^{-1} = v_i^2$ are replaced by

$$tv_it^{-1} = v_i^m \quad (1 \leq i \leq 3);$$

thus $E = E_2$, while the marked word $w_m = [tct^{-1}, v_1(tct^{-1})v_1^{-1}]$ is unchanged.

Theorem 9.15 (scaling family). *For every $m \geq 2$: the group E_m is finitely presented, w_m is a nontrivial central involution in E_m , and every corona representation of E_m kills w_m . Consequently E_m is not MF, neither $C_{\text{max}}^*(E_m)$ nor $C_{\text{red}}^*(E_m)$ is an MF algebra, and $C_{\text{red}}^*(E_m)$ is stably finite but not MF.*

Proof. Finite presentation is clear from the displayed presentation.

The pattern. Define $\alpha_m: \Gamma \rightarrow \Gamma$ by $\alpha_m(v, A) = (mv, A)$. The computation of Lemma 3.1 applies with 2 replaced by m : α_m is an injective endomorphism with image $m\mathbb{Z}^3 \rtimes \text{SL}_3(\mathbb{Z})$, and $a = (e_1, 1) \notin \alpha_m(\Gamma)$ because $e_1 \notin m\mathbb{Z}^3$. On the six base generators α_m fixes x, y, z and sends v_i to v_i^m , so the relations of E_m assert $t\gamma t^{-1} = \alpha_m(\gamma)$ on generators; hence, with $\iota_m: \Gamma \rightarrow E_m$ the canonical homomorphism, $t\iota_m(\gamma)t^{-1} = \iota_m(\alpha_m(\gamma))$ for all $\gamma \in \Gamma$, and c commutes with $\iota_m(\Gamma)$: the tuple (Γ, ι_m, t, c) is a marked Kazhdan pattern in E_m (Fact 2.1). As in the proof of Theorem A, $d^2 = tc^2t^{-1} = 1$ in E_m , and Lemma 8.1 gives $w_m = [d, \iota_m(a)]^2 \in N_{\text{comp}}$; the subgroup $\{1, w_m\}$ is finite and normal, so Theorem 7.2 kills w_m in every corona representation.

Nontriviality. Let $D_m = \text{diag}(m, m, m, 1) \in \text{GL}_4(\mathbb{Z}[1/m])$. As in Lemma 4.2, $D_m \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} D_m^{-1} = \begin{pmatrix} A & mv \\ 0 & 1 \end{pmatrix}$ is the matrix of $\alpha_m(v, A)$, and $D_m^{-1}aD_m = \begin{pmatrix} 1 & \frac{1}{m}e_1 \\ 0 & 1 \end{pmatrix} \notin \Gamma$, so in $\bar{G}_m := \langle \Gamma, D_m \rangle \leq \text{GL}_4(\mathbb{Z}[1/m])$ the cosets $D_m\Gamma$ and $aD_m\Gamma$ of $\bar{X}_m := \bar{G}_m/\Gamma$ are distinct. The proof of Proposition 4.3 applies with $(\bar{G}, \bar{X}, D)$ replaced by $(\bar{G}_m, \bar{X}_m, D_m)$ — its only inputs were the conjugation identity, the fixing of the base coset by Γ , and the coset separation — and sends w_m to $[c_{D_m\Gamma}, c_{aD_m\Gamma}] = -1 \neq 1$ in $\text{Cilamp}(\bar{X}_m) \rtimes \bar{G}_m$.

The final assertions follow exactly as in the proofs of Theorems A and D, with E_m in place of E . $\square$

Remark 9.16. We do not claim the groups E_m are pairwise non-isomorphic; multiplicity is addressed in Corollary 9.18. For every $d \geq 3$ the same construction runs with $\mathbb{Z}^d \rtimes \mathrm{SL}_d(\mathbb{Z})$ in place of Γ — property (T) holds by the same references [BHV, Sections 1.4–1.5 and 4.2] — once a finite presentation of the base is fixed; we confine the formal statement to $d = 3$, where the presentation (3) is completely explicit.

9.7. Multiplicity.

Lemma 9.17. *Every subgroup of an MF group is MF; equivalently, a group containing a non-MF subgroup is not MF.*

Proof. The restriction of an injective corona representation to a subgroup is an injective corona representation. $\square$

Corollary 9.18 (multiplicity).

- (1) *The groups $E \times \mathbb{Z}^k$, $k \in \mathbb{N}$, are pairwise non-isomorphic, finitely presented, and non-MF.*
- (2) *There exist $2^{\aleph_0}$ pairwise non-isomorphic finitely generated non-MF groups.*

Proof. Every group in sight contains an isomorphic copy of E , hence is non-MF by Theorem A and Lemma 9.17.

(1) $E \times \mathbb{Z}^k$ is finitely presented because E is. Its abelianization is $E^{\mathrm{ab}} \times \mathbb{Z}^k$; since E is finitely generated, E^{ab} is a finitely generated abelian group, of some torsion-free rank $r_0 < \infty$, so the abelianization of $E \times \mathbb{Z}^k$ has torsion-free rank $r_0 + k$, an isomorphism invariant that separates distinct k .

(2) By a classical construction of B. H. Neumann [Neu] there is a family $(N_i)_{i \in I}$ with $|I| = 2^{\aleph_0}$ of pairwise non-isomorphic 2-generator groups. Put $G_i = E \times N_i$, a finitely generated non-MF group. For a fixed countable group G_1 , the set $\{i \in I : G_i \cong G_1\}$ is countable: a countable group has only countably many finitely generated subgroups (each is determined by a finite generating subset), hence countably many isomorphism classes of such; if $G_i \cong G_1$ then N_i embeds into G_1 as $\{1\} \times N_i$; and the N_i are pairwise non-isomorphic, so $i \mapsto [N_i]$ is injective on the set in question. A map from a set of cardinality $2^{\aleph_0}$ with countable fibres has image of cardinality $2^{\aleph_0}$; apply this to $i \mapsto [G_i]$. $\square$

9.8. Uniform invisibility.

Theorem 9.19 (uniform invisibility). *Let H , (Λ, ι, t, c) and $F \subseteq N_{\mathrm{comp}}$ be as in Theorem 7.2, and fix $f_0 \in F$ — for instance $H = E$ and $f_0 = w$. For every $\varepsilon > 0$ there exist $\delta > 0$ and a finite subset $F_0 \subseteq H$ such that for every $r \geq 1$ and every map $\varphi : H \rightarrow \mathcal{U}(M_r(\mathbb{C}))$ with*

$$\|\varphi(g)\varphi(h) - \varphi(gh)\| \leq \delta \quad \text{for all } g, h \in F_0,$$

one has $\|\varphi(f_0) - 1\| < \varepsilon$.

Formal counterpart:

`GroupApproximation.KazhdanCompressionCore.finiteNormal_uniform_invisibility.`

Proof. Suppose the statement fails for some $\varepsilon_0 > 0$. Fix an enumeration $H = \{h_1, h_2, \dots\}$ and put $F_k = \{h_1, \dots, h_k\}$. For each $k \geq 1$, the failure at $(\delta, F_0) = (1/k, F_k)$ produces $r_k \geq 1$ and $\varphi_k: H \rightarrow \mathcal{U}(M_{r_k}(\mathbb{C}))$ with $\|\varphi_k(g)\varphi_k(h) - \varphi_k(gh)\| \leq 1/k$ for all $g, h \in F_k$, but $\|\varphi_k(f_0) - 1\| \geq \varepsilon_0$. Define $\Theta(g)$ as the class of the sequence $(\varphi_k(g))_{k \in \mathbb{N}}$ in $\mathcal{Q}((r_k)_{k \in \mathbb{N}})$, a unitary element. For fixed $g, h \in H$ we have $g, h \in F_k$ for all large k , so the displayed defect tends to 0 and $\Theta(g)\Theta(h) = \Theta(gh)$. Taking $g = h = 1$ gives $\|\varphi_k(1)^2 - \varphi_k(1)\| \leq 1/k$ for large k , hence $\|\varphi_k(1) - 1\| \leq 1/k$ after multiplying by the unitary $\varphi_k(1)^*$; so $\Theta(1) = 1$, and then $\Theta(g)\Theta(g^{-1}) = \Theta(1) = 1$ for every g . Thus Θ is a corona representation of H , and Theorem 7.2 gives $\Theta(f_0) = 1$, i.e. $\|\varphi_k(f_0) - 1\| \rightarrow 0$ — contradicting $\|\varphi_k(f_0) - 1\| \geq \varepsilon_0$. $\square$

Remark 9.20. Conversely, the uniform statement applied to unitary lifts recovers the kill part of Theorem 7.2, so the two are equivalent. The compactness proof yields no explicit (δ, F_0) , and none is claimed; an effective version would require tracking the Kazhdan constant of Λ through Sections 7–8, which we do not pursue.

Remark 9.21 (open questions). Three questions adjacent to this section. (i) Is $E/\langle w \rangle$ MF, i.e. is the MF radical of E exactly $\{1, w\}$? (ii) Does a *torsion-free* non-MF group exist? The mechanism here is torsion-critical: the criterion kills a finite normal subgroup, which a torsion-free group cannot supply. The analogous sofic-side emphasis on torsion appears in Fournier-Facio’s torsion-free nonsofic group [FF]. (iii) Is non-MF-ness an open condition in the space of marked groups, as soficity questions are studied there [FF]?

10. THE CYCLIC-BASE COMPARISON

We prove Theorem C. The broader phenomenon that a marked central involution can be invisible to every exact finite-dimensional representation but visible to approximate representations is central to Slofstra’s work [Slo17]; the point here is its occurrence in the same operator-norm marked-compression pattern as Theorem A. Recall the setting: $G_0 = \text{BS}(1, 2)$ with base $\Gamma_0 = \langle \gamma_0 \rangle \cong \mathbb{Z}$ and stable letter t satisfying $t\gamma_0 t^{-1} = \gamma_0^2$; $X_0 = G_0/\Gamma_0$; $E_0 = \text{CLamp}(X_0) \rtimes G_0$ with central sign $z_0 = -1 \in \text{CLamp}(X_0)$ as in Construction 4.1.

Proof of Theorem C. $z_0 \neq 1$: immediate from Construction 4.1, since $\text{CLamp}(X_0) \leq \text{Cl}(X_0)^\times$ and $\text{Cl}(X_0) \neq 0$.

Amenability: every finitely generated subgroup of $\text{CLamp}(X_0)$ is contained in the subgroup generated by ± 1 and finitely many c_x , which is a finite 2-group (it has order at most 2^{k+1} for k sites, since every element is $\pm a$ a reduced product of distinct generators). Hence $\text{CLamp}(X_0)$ is locally finite, so amenable. The group $G_0 = \text{BS}(1, 2) \cong \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ is metabelian, so

amenable. As an extension of the amenable G_0 by the amenable $\text{Cilamp}(X_0)$, the group E_0 is amenable.

MF: by Facts 2.3 and 2.4, the amenable group E_0 is MF, and an MF embedding $E_0 \hookrightarrow \mathcal{U}(\mathcal{Q})$ is injective, so the image of z_0 is a nontrivial (self-adjoint) unitary in the corona: the central sign survives.

The quotient from the literal presentation: assign $\gamma_0 \mapsto \gamma_0$, $t \mapsto t$, and $c \mapsto c_o$, where $o = \Gamma_0 \in X_0$. The point o is fixed by Γ_0 , so $[c_o, \gamma_0] = 1$. Also $t\Gamma_0 t^{-1} = \langle \gamma_0^2 \rangle \subseteq \Gamma_0$. Moreover $\gamma_0 t \Gamma_0 \neq t \Gamma_0$ in G_0 : otherwise $t^{-1} \gamma_0 t \in \Gamma_0$; in the faithful affine model $\text{BS}(1, 2) = \mathbb{Z}[1/2] \rtimes \mathbb{Z}$ with $\gamma_0 = (1, 0)$ and $t = (0, 1)$ acting by multiplication by 2, we compute $t^{-1} \gamma_0 t = (\frac{1}{2}, 0) \notin \Gamma_0 = \mathbb{Z} \times \{0\}$. Consequently

$$w_{\text{BS}} \mapsto [c_{t\Gamma_0}, c_{\gamma_0 t \Gamma_0}] = z_0.$$

The image is central of order two, so every relator of E_{BS} is respected. The resulting homomorphism is surjective: γ_0, t generate G_0 , the conjugates of c_o give all c_x because G_0 acts transitively on X_0 , and their commutators contain z_0 . Composing this quotient with the MF embedding of E_0 gives the asserted corona representation of E_{BS} .

The finite-dimensional obstruction persists: E_0 contains the pattern of Theorem B with $H = E_0$, $\Gamma = \Gamma_0$, $t = t$, $c = c_o$ ($o = \Gamma_0 \in X_0$, fixed by Γ_0), and $a = \gamma_0 \in \Gamma_0$. We have $t\Gamma_0 t^{-1} = \langle \gamma_0^2 \rangle \subseteq \Gamma_0$ and $[c_o, \Gamma_0] = 1$. Thus Theorem B says every finite-dimensional representation of E_0 , over every field, kills z_0 . Applying the same theorem directly to the displayed presentation of E_{BS} shows that every finite-dimensional representation kills w_{BS} . $\square$

Remark 10.1. Theorem C shows that the argument cannot be generalized merely by deleting its fixed-space rigidity input. The elementary Theorem B is genuinely weaker than the corona conclusion: E_0 has MF approximations even though its finite-dimensional representations all kill the sign.

11. LIMITATIONS: HILBERT–SCHMIDT METRICS AND HYPERLINEARITY

The proof of Theorem A does not extend to tracial ultraproducts, for a structural reason worth recording.

Suppose one replaces the norm corona $\mathcal{Q}$ by a tracial ultraproduct $\mathcal{M} = \prod^\omega (M_{d_n}(\mathbb{C}), \text{tr}_{d_n})$, with defects measured in $\|\cdot\|_2$ rather than operator norm, and asks whether a hyperlinear-style embedding of E must kill w . Two steps break.

First, Lemma 7.4 fails: Hilbert–Schmidt-multiplicative lifts need not be operator-norm multiplicative, so the adjoint maps $\text{Ad } U_{g,n}$ need not assemble into a homomorphism into any norm corona. One can instead pass to the conjugation representation on $L^2(\mathcal{M})$, which is an honest representation of E .

The tracial ultraproduct $\mathcal{M}$ itself is a finite von Neumann algebra. Second, and decisively, Lemma 6.2 is unavailable after the argument passes to its conjugation action. The available ambient algebra is $B(L^2(\mathcal{M}))$; when

$L^2(\mathcal{M})$ is infinite-dimensional this algebra is properly infinite and has no normalized finite trace with which to compare the fixed-space projections. Thus the inclusion supplied by the analogue of Lemma 7.5 is not forced to be equality. Proper inclusion between nested unitarily equivalent projections is possible in such an ambient algebra (the unilateral-shift phenomenon), but we do not claim that it occurs for a tracial representation of the present group. In particular the present paper proves nothing about hyperlinearity or Connes embeddability of E , and nothing about the sofic-versus-hyperlinear problem.

The dividing line in the present argument is thus: *operator-norm matrix models form a stably finite world, in which property (T) compression is rigid; for tracial matrix models the available ambient algebra of the conjugation action can be properly infinite, so the same projection comparison need not collapse to equality.*

This failure is closely related to a current structural question rather than to a known impossibility theorem. Alekseev–Thom ask whether, for a Kazhdan representation into a tracial ultraproduct of matrices, its commutant can be recovered as an ultraproduct of finite-dimensional subalgebras, possibly as coordinate centralizers of suitable lifts [AT, Open Problem 6.2]. Such a coordinate model would provide finite-level structure absent from the argument above, but their question is open and does not restore the present stable-finiteness proof in the tracial setting.

TOOL AND COMPUTATIONAL RESOURCE DISCLOSURE

The mathematics in this paper was developed and formalized with extensive use of large language models, including Anthropic Claude and OpenAI Codex models in multiple concurrent interactive sessions, operating under continuous human direction. The formal counterpart cited in the Lean-status remarks was produced the same way and is verified by the Lean 4 kernel against Mathlib, with the axiom closure and audit apparatus described in the repository. Builds and audits ran on the Minnesota Supercomputing Institute cluster. Literature search was assisted by web-connected model sessions; every attribution-critical claim was verified against primary sources by direct reading. The author retains exclusive responsibility for the correctness and adequacy of the arguments and for the completeness and accuracy of the citations, in line with the Leiden Declaration on Artificial Intelligence and Mathematics (2026).

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